

The Cobordism Functor, the Theta Register, and Root Systems

Derivation and experiment specification

Abstract

This document specifies the harmonic-operator experiment: its analytical derivation, geometric operator algebra, theta tensor register, and root-system extension. The objective is an exact representation of finite-dimensional algebra by quantized geometry, not synthesis of every operator as one engine-generated spacetime. Harmonic forms parameterize a classical phase space; quantization supplies the quantum Hilbert space. Geometric translations and their complex linear combinations form its full matrix algebra, preserving tensor products, adjoints, states and operator vectors. Root-lattice geometry represents characters and fusion with explicitly declared quantization data. Single-cobordism synthesis and physical dynamics are separate questions. Theorems, measurements and open constructions are distinguished throughout.

Scope and notation

The source is `docs/design/cobordism_functor_reading.tex` in the Tessera repository; the adjacent PDF is generated from it, and the copies in `~/tessera-notes/cobordism-functor/` are reading copies. Open work is tracked on GitHub, not here. The companion map, problems (P1–P83) and solutions are study aids; where they conflict with this document, this document governs.

Notation. D is geometric dimension, p cochain degree, r_H harmonic dimension, g the dimension of a polarized Jacobian, and r the rank of a Lie algebra. The primer uses k for cochain degree in its Hodge discussion and for the positive integer quantization level in its theta/affine discussion; those are independent quantities. In $\mathfrak{su}(N)$, N is the defining representation dimension, not simplex dimension. M denotes period transport; $\mathbf{M}$ denotes its standard Siegel-coordinate counterpart where that bridge applies; M_p denotes a mass matrix. Ω_∂ is the boundary intersection form; Ω is a complex period matrix. Transpose and Hermitian adjoint are never interchangeable without the stated real or duality hypotheses.

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Part I

Primer

1 What is being claimed

Let W be a compact oriented triangulated 3-manifold whose boundary is two tori, $\partial W = M_0 \sqcup M_1$. Each torus carries a *marking*: two cycles A, B that form a basis of $H_1(M_i; \mathbb{Z}) \cong \mathbb{Z}^2$; their periods give the dual cohomology coordinates. The claim has three levels. The goal is an equivalent geometric representation of algebra, including complex linear combinations of geometric operations. Bare geometry is not claimed to choose a state, level or physical dynamics.

1. (*Exact.*) The degree-one harmonic space of W , restricted to the boundary, is a subspace $L_W \subset H^1(M_0) \oplus H^1(M_1)$ that is Lagrangian for the boundary intersection form and, when it is the graph of a map, defines a metric-independent matrix M . If the restrictions identify the full integral lattices, M is integral and unimodular. Gluing cobordisms multiplies their matrices. M does not depend on the metric.
2. (*Exact, on a new register.*) Quantizing each marking lattice at level k gives a k -dimensional space Θ_k per torus; a disjoint union of tori gives the tensor product; an appropriately lifted symplectic period map acts by the unitary projective matrix $\rho_k(M)$ after the coordinate bridge, with a declared polarization, theta structure and lift convention. Geometric clock and shift operations span the full operator algebra (Theorem 12.2), not only the modular gates. At even level, including $k = 2$, these operators are Clifford; antisymplectic maps require an antiunitary interpretation.
3. (*Exact representation with declared root data.*) Replacing the abelian lattice by a root system gives the non-abelian register: Weyl-antisymmetrized theta functions, Kac–Peterson modular matrices, Verlinde fusion, and — with punctures — braid representations with proven density in specified sectors. An internal scalar Euclidean lattice realizes A_2 explicitly (Proposition 18.3). Level and affine reduction are declared structure.

For fixed topology, markings and flat local system, the metric controls harmonic representatives, pairings and the validity of the harmonic model. It also affects metric-dependent observations and dynamics. None of this makes the period relation itself metric-dependent.

Definition 1.1 (Representation versus synthesis). A geometric representation is a geometric vector space or algebra with an invertible map to its algebraic counterpart preserving specified operations: here addition, complex scalar multiplication, composition, adjoint and tensor product. Symbols may be complex linear combinations of decorated cycles; their coefficients carry information and need not emerge from a metric. Synthesis asks additionally for one allowed scalar simplex complex whose native readout realizes a specified operator. The experiment requires the first statement, not the second.

2 Cochains, the Laplacian, and the harmonic space

A k -chain is a formal integer combination of oriented k -simplices; ∂_k is the boundary map, $\partial_{k-1}\partial_k = 0$. A k -cochain is a linear function on k -chains: one number per oriented simplex, changing sign with orientation. The coboundary $d_k = \partial_{k+1}^\top$ satisfies $\langle d\omega, c \rangle = \langle \omega, \partial c \rangle$ and $d_{k+1}d_k = 0$. The period of a cochain over a cycle z ($\partial z = 0$) is $\langle \omega, z \rangle$.

Lemma 2.1. A coboundary has zero periods: $\langle d\eta, z \rangle = \langle \eta, \partial z \rangle = 0$ for every cycle z .

Cohomology $H^k = \ker d_k / \operatorname{im} d_{k-1}$ is the quotient of closed cochains by coboundaries; its dimension is the Betti number b_k .

A metric enters through positive matrices M_k on k -cochains (for us the Whitney mass, a function of the edge lengths) and gives the codifferential $\delta_k = M_{k-1}^{-1} d_{k-1}^\dagger M_k$ and the Hodge Laplacian $L_k = d_{k-1} \delta_k + \delta_{k+1} d_k$. The *harmonic space* is $\mathcal{H}_k = \ker L_k = \ker d_k \cap \ker \delta_k$.

Theorem 2.2 (Discrete Hodge). *If the cochain masses are positive Hermitian then $C^k = \operatorname{im} d_{k-1} \oplus \mathcal{H}_k \oplus \operatorname{im} \delta_{k+1}$, orthogonally, and $\mathcal{H}_k \rightarrow H^k$ (a harmonic cochain to its class) is an isomorphism. In particular $\dim \mathcal{H}_k = b_k$ and every class has exactly one harmonic representative.*

Remark 2.3 (The Lorentzian caveat). On complex lengths M_k is complex bilinear and not positive, and the positive Hodge theorem does not apply. Equality of dimensions alone does not imply a harmonic identification with cohomology. Certify closedness, coclosedness, an independent kernel of the same span, and that the map to cohomology is injective and onto (for example by a full period-rank test). The code's H1–H3 checks form part of this contract; see Section 26. All uses of harmonic representatives in the cohomological theorems assume this identification. Quantum statements add a positive Hermitian Hilbert metric explicitly.

Theorem 2.4 (Periods do not move). *Fix a class $\alpha \in H^k$ and a cycle z . If ω and ω' are the harmonic representatives of α for two metrics, then $\langle \omega, z \rangle = \langle \omega', z \rangle$.*

Proof. Both represent α , so $\omega - \omega' = d\eta$; apply Lemma 2.1. □

3 Restriction to the boundary

Write $\iota : \partial W \hookrightarrow W$ for the inclusion and $\iota^* : H^k(W) \rightarrow H^k(\partial W)$ for restriction of classes. On cochains this restricts values to boundary cells. A closed cochain restricts to a closed cochain, but generally not to a coclosed one for the boundary's own metric. Passing to classes, or then taking the boundary harmonic representative when defined, is essential. Let $m = \dim \partial W$ ($m = 2$ for us).

Theorem 3.1 (Mutual annihilators). *Under the Poincaré pairing $H^k(\partial W) \times H^{m-k}(\partial W) \rightarrow \mathbb{R}$,*

$$\operatorname{im} (H^k(W) \rightarrow H^k(\partial W)) = \operatorname{ann} \left(\operatorname{im} (H^{m-k}(W) \rightarrow H^{m-k}(\partial W)) \right).$$

The proof has two halves. Orthogonality is Stokes with the cup product: for closed $\alpha \in C^k(W)$ and $\beta \in C^{m-k}(W)$,

$$\langle \alpha|_{\partial W} \cup \beta|_{\partial W}, [\partial W] \rangle = \langle \alpha \cup \beta, \partial[W] \rangle = \langle d(\alpha \cup \beta), [W] \rangle = 0,$$

since $d(\alpha \cup \beta) = d\alpha \cup \beta \pm \alpha \cup d\beta = 0$. That the two images are *exact* annihilators, not merely orthogonal, is a dimension count from the long exact sequence of the pair $(W, \partial W)$ together with Lefschetz duality $H^k(W, \partial W) \cong H_{m+1-k}(W)$. Both halves hold for compact oriented triangulated manifolds over a field; arbitrary simplicial complexes need not satisfy Poincaré–Lefschetz duality. No metric or signature appears.

Corollary 3.2 (Half lives, half dies). *For ∂W a union of tori and $k = 1$: $\operatorname{rank}(H^1(W) \rightarrow H^1(\partial W)) = \frac{1}{2} b_1(\partial W)$, and the image is a Lagrangian subspace for the intersection form on $H^1(\partial W)$.*

For two tori, $b_1(\partial W) = 4$ and exactly two of the four boundary periods can be prescribed; the bulk decides which two. Call the image L_W . It is a linear *relation* between $H^1(M_0)$ and $H^1(M_1)$.

Remark 3.3 (The orientation sign). Write the intersection form on one torus in its marking as $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The boundary form is $\Omega_\partial = J_A \oplus \varepsilon J_B$ where $\varepsilon = \pm 1$ records that the two ends inherit *opposite* orientations from the bulk, corrected by the orientation of any relabelling φ applied to the far surface: $\varepsilon = -\det \varphi^*$. This sign is declared data, never searched for. Problem P6 shows why it cannot be read off $\det M$ once there are four tori. The displayed rule applies to the declared collar markings. With more components, assemble one signed block per boundary component from its actual induced orientation; a total determinant is insufficient.

4 The monodromy

Let Z be a basis of $\mathcal{H}_1(W)$ and let P_A, P_B be the matrices of periods of Z over the two markings ($2 \times r_H$ each). When P_A is invertible, L_W is the graph of the map

$$M = P_B P_A^{-1}, \quad p_{\text{out}} = M p_{\text{in}}.$$

Proposition 4.1. *M does not depend on the basis Z : a change $Z \mapsto ZS$ sends $P_A \mapsto P_AS$, $P_B \mapsto P_BS$, and $P_BS(P_AS)^{-1} = P_BP_A^{-1}$.*

Theorem 4.2 (Integrality under integral lattice identification). *If the two restrictions identify the bulk lattice (modulo invisible classes and torsion) with each full boundary integral lattice, then $M \in \text{GL}(2, \mathbb{Z})$: it is an integer matrix with $\det M = \pm 1$.*

Proof. M is, by this additional hypothesis, the matrix of a $\mathbb{Z}$ -module isomorphism between the two rank-two lattices, hence integer with a unit determinant. $\square$

Corollary 4.3. *No change of metric changes M : it was defined without one.*

An invertible P_A over $\mathbb{C}$ alone does not prove the integral hypothesis. For example, $P_A = \text{diag}(1, 2)$ and $P_B = \text{diag}(2, 1)$ give $M = \text{diag}(2, 1/2)$, a rational symplectic matrix, not an integral one. This is an algebraic counterexample to the inference, not a manifold constructed here. Mapping cylinders satisfy the integral-isomorphism hypothesis.

For a product collar $T^2 \times [0, 1]$, $M = I$. Relabelling the far end by a simplicial automorphism φ of the torus gives $M = \varphi^*$. On the square-grid triangulation with one diagonal an orientation-reversing such φ is the swap $(i, j) \mapsto (j, i)$, with $\varphi^* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\det = -1$.

5 From a graph relation to an operator vector

For finite-dimensional A, B , the Choi–Jamiołkowski identification is $\text{Hom}(A, B) \cong B \otimes A^*$: a map T corresponds to the vector

$$|T\rangle\rangle = \sum_i T e_i \otimes e_i^*,$$

and its graph $\{(x, Tx)\} \subset A \oplus B$ is a subspace of dimension $\dim A$. Conversely a subspace of $A \oplus B$ that meets $0 \oplus B$ trivially and projects onto A is the graph of exactly one map.

Thus $L_W \leftrightarrow M \leftrightarrow |M\rangle\rangle$ is an exact correspondence *when the graph conditions hold*; it is not literal equality of a harmonic space, a graph subspace and a vector. The dual input factor must not be dropped. In Hilbert notation it becomes a conjugate input factor, in the same output-major order as Section 29. A nonzero output with zero input violates uniqueness; an input which cannot extend violates coverage. A map taking a nonzero input to zero is still a valid graph and need not be invertible.

6 Composition: gluing is matrix multiplication

Let $W = W_1 \cup_{M_1} W_2$ with $W_1 : M_0 \rightarrow M_1$ and $W_2 : M_1 \rightarrow M_2$.

Theorem 6.1. $L_W = L_{W_2} \circ L_{W_1}$ as relations; when both are graphs, $M(W) = M(W_2) M(W_1)$. The cylinder $M \times [0, 1]$ gives $M = I$, and a disjoint union gives the direct sum.

Proof sketch. Mayer–Vietoris for $W = W_1 \cup W_2$ with $W_1 \cap W_2 = M_1$: a cohomology class on W restricts to classes on W_1 and W_2 that agree in $H^1(M_1)$, and exactness says each such compatible pair extends. The extension need not be unique: the connecting map from $H^0(M_1)$ can contribute invisible classes. Restricted harmonic representatives need not themselves be harmonic. Compatibility of classes gives exactly the composition of external boundary relations. (On a simplicial complex the glued object must actually be a complex — one cell structure on the seam — or the sequence does not apply.) $\square$

This gives a functor to linear relations, with disjoint union mapped to direct sum; on the graph subcategory it gives linear maps. It is not by itself an Atiyah vector-space TQFT, whose monoidal operation is tensor product. Theta factorization supplies tensor spaces on objects, but a full quantization of every cobordism relation requires further construction.

7 What the metric controls

For the cohomological correspondence the relevant distinctions are:

1. Whether the harmonic space actually identifies with cohomology — automatic for positive Hodge metrics, a certificate otherwise.
2. The boundary pairings, hence whether M preserves them. Green’s formula gives the *annihilator* pairing (tangential against normal), not an isometry: nothing in Sections 3–6 makes M unitary. Dual/primal bilinear transport and positive Hermitian quantum isometry require distinct tests and explicit coordinate conventions (Section 27).
3. Which cochain in each class is harmonic, and the frame it is read in. Theorem 2.4: the representative moves, the periods do not.

Remark 7.1 (Gauge covariance of the pairing). For scalar local system U , use its dual U^{-1} in the bilinear pairing. If primal and dual coordinates transform by ρ and $\tilde{\rho}$, then the dressed mass transforms as $G' = \tilde{\rho}^{-\top} G \rho^{-1}$, so $(\tilde{\rho} \tilde{c})^\top G' (\rho c) = \tilde{c}^\top G c$. Pairing two primal frames is not a substitute. An own-metric biorthogonal frame can satisfy $\tilde{\Phi}^\top M_{\text{own}} \Phi = I$; that does not make a whole-metric observation an orthogonal projection, nor make its period-coordinate Gram the identity. See Section 27.

Proposition 7.2 (Pinned-boundary defect is invariant). *Fix own-boundary pairings G_A, G_B , coordinate markings and primal/dual period transports $M, M^\vee$. Then*

$$D = \frac{\|(M^\vee)^\top G_B M - G_A\|_F}{\|G_A\|_F}$$

is independent of interior metric changes, including relaxation.

Proof. Every argument of the displayed function is fixed. Own-boundary Grams depend on boundary data, not the bulk metric. At zero connection primal and dual transports coincide; otherwise this requires proof. $\square$

8 Weights add, characters multiply

For a torus group $T = \mathfrak{h}/\Lambda$ a representation decomposes into characters $e^{2\pi i \langle \mu, \cdot \rangle}$ labelled by *weights* $\mu \in \Lambda^*$. Characters multiply under tensor product; their weights *add*: the weights of $V \otimes V'$ are the sums of a weight of V and a weight of V' . And the weight lattice of a product torus $T_1 \times T_2$ is $\Lambda_1^* \oplus \Lambda_2^*$. So

direct sum of weight lattices $\longleftrightarrow$ tensor product of representations.

This motivates, but does not construct, the finite theta register. An integral symplectic lattice, a positive polarization and a level line bundle define $H^0(\text{Jac}(M), L^k)$; all lattice characters alone would not give this finite-dimensional Hilbert space. Products of polarized Jacobians with external-product line bundles give tensor products of section spaces. Lie algebras (Sections 18–20) handle the non-abelian case, where adding weights must be followed by reflecting back into a fundamental domain of the Weyl group (Proposition 19.2).

9 Harmonic phase space and scalar quantization

Definition 9.1 (Flat phase configuration and Jacobian). On a closed oriented genus- g surface Σ , let ϕ be a real ordinary closed 1-cochain. Gauge transformations add $d\chi$; large gauge transformations shift its periods by $2\pi\mathbb{Z}$. Define

$$a_i = \frac{1}{2\pi} \oint_{A_i} \phi, \quad b_i = \frac{1}{2\pi} \oint_{B_i} \phi, \quad J(\Sigma) = H^1(\Sigma; \mathbb{R})/H^1(\Sigma; \mathbb{Z}).$$

The Wilson loops are $e^{2\pi i a_i}, e^{2\pi i b_i}$; a real lift is coordinate data, not another observable. Positive Hodge masses select a unique harmonic representative of each real cohomology class. The integral lattice and intersection pairing give the principal polarization, represented by $\sum_i da_i \wedge db_i$.

For symmetric Ω with $\text{Im } \Omega > 0$, use

$$z = b - \Omega a \pmod{\mathbb{Z}^g + \Omega \mathbb{Z}^g}.$$

This fixes the real-holonomy to complex-Jacobian coordinate map. Positive cochain masses prove Hodge decomposition but do not alone prove that a discretized star yields a Riemann period matrix; certify the complex structure or establish a controlled approximation (Section 30).

Configuration versus charged differential. Here ϕ is a configuration variable and d remains the ordinary coboundary. It is not inserted into d_U to define a charged harmonic register. On a torus with holonomies u, v , the minimal cellular differentials of the rank-one local system are

$$d_0 = \begin{pmatrix} u-1 \\ v-1 \end{pmatrix}, \quad d_1 = \begin{pmatrix} -(v-1) & u-1 \end{pmatrix}.$$

For $(u, v) \neq (1, 1)$ both have rank one and all cohomology groups vanish. Thus twisting the pencil with the state-selection phases changes the problem. Store/read phase configurations separately from neutral Hodge assembly; reuse scalar Wilson-loop transport without mutating that assembly.

Definition 9.2 (Quantized register). Choose an integer $k > 0$, a positive polarization and a theta structure (a compatible lift of the finite translations to a line bundle). The register is $\mathcal{Q}_k(\Sigma) = H^0(J(\Sigma), L^k)$ with its positive section norm, where L^k is the level- k line bundle. It is a quantization of harmonic phase space, not equality with $\ker \Delta_1$. Its line connection is scalar and lives over the Jacobian, with curvature encoding the polarization.

10 Theta functions: the register on one torus

Let τ lie in the upper half-plane and $\Lambda = \mathbb{Z} + \tau\mathbb{Z}$. The Jacobian $\mathbb{C}/\Lambda$ is the torus itself, and a marking $(A, B) = (1, \tau)$.

Definition 10.1 (Level- k theta functions). For $j \in \mathbb{Z}/k$,

$$\theta_j(z; \tau) = \sum_{n \in \mathbb{Z}} \exp\left(\pi i k \left(n + \frac{j}{k}\right)^2 \tau + 2\pi i k \left(n + \frac{j}{k}\right) z\right).$$

$\Theta_k(\tau) = \text{span}\{\theta_0, \dots, \theta_{k-1}\}$ is the register's Hilbert space for one torus; $k = 2$ is a qubit.

The sum converges absolutely because $\text{Im } \tau > 0$. Each θ_j is periodic under $z \mapsto z + 1$ and *quasi*-periodic under $z \mapsto z + \tau$:

$$\theta_j(z + \tau; \tau) = e^{-\pi i k \tau - 2\pi i k z} \theta_j(z; \tau),$$

which makes θ_j a section of a line bundle of degree k over the torus rather than a function. The quantity $\|\theta_j\|(z) = |\theta_j(z; \tau)| e^{-\pi k (\text{Im } z)^2 / \text{Im } \tau}$ is doubly periodic (Problem P21), and each θ_j has exactly k zeros in a fundamental domain (Problem P22).

Translations by the k -torsion points act on the basis:

$$\theta_j\left(z + \frac{1}{k}\right) = e^{2\pi i j/k} \theta_j(z), \quad \theta_j\left(z + \frac{\tau}{k}\right) = e^{-\pi i \tau/k - 2\pi i z} \theta_{j+1}(z),$$

the *clock* and *shift* operators Z, X of a k -level system, with $ZX = e^{2\pi i/k} XZ$. They generate the Heisenberg–Weyl group of the torus's k -torsion. These are operations on sections. Matrices on a column of evaluated basis functions and on section coefficients are transposes of one another; coherent-state transport is fixed in Section 15.

11 Direct sum to tensor product

For g tori take a period matrix Ω : $g \times g$, symmetric, with positive-definite imaginary part (the Siegel upper half-space). For a disjoint union $\Omega = \text{diag}(\tau_1, \dots, \tau_g)$. With characteristics $j \in (\mathbb{Z}/k)^g$,

$$\theta_j(z; \Omega) = \sum_{n \in \mathbb{Z}^g} \exp\left(\pi i k \left(n + \frac{j}{k}\right)^T \Omega \left(n + \frac{j}{k}\right) + 2\pi i k \left(n + \frac{j}{k}\right)^T z\right).$$

Proposition 11.1. *If $\Omega = \text{diag}(\tau_1, \tau_2)$ then $\theta_{(j_1, j_2)}(z; \Omega) = \theta_{j_1}(z_1; \tau_1) \theta_{j_2}(z_2; \tau_2)$, so $\Theta_k(\tau_1 \oplus \tau_2) = \Theta_k(\tau_1) \otimes \Theta_k(\tau_2)$.*

Proof. The quadratic form and the linear term both split, so the sum over $\mathbb{Z}^2$ is a product of sums over $\mathbb{Z}$. $\square$

Theorem 11.2 (Dimension and positive section metric). *For a principally polarized g -dimensional complex torus and $k > 0$, the theta functions indexed by $j \in (\mathbb{Z}/k)^g$ form a basis of $\mathcal{Q}_k$, of dimension $d = k^g$. Put $Y = \text{Im } \Omega$ and $z = x + \Omega y$. The positive Hermitian norm making this basis orthonormal is*

$$\langle f, h \rangle = (2k)^{g/2} \sqrt{\det Y} \int_{[0,1]^{2g}} \overline{f(x + \Omega y)} h(x + \Omega y) e^{-2\pi k y^T Y y} dx dy.$$

Proof. Periodicity in x gives a Fourier series. The Ω quasi-periodicity determines all coefficients from their k^g residue classes modulo k ; the theta series realize those independent choices. Integration in x makes distinct frequencies orthogonal. Summing the translates of the y cube gives the Gaussian integral $(2k)^{-g/2} (\det Y)^{-1/2}$, canceled by the stated normalization. $\square$

12 The full geometric operator algebra

Definition 12.1 (Reduced geometric translation algebra). Let $\omega = e^{2\pi i/k}$ and $p, q \in (\mathbb{Z}/k)^g$. The complex vector space $\mathcal{A}_{g,k}$ has basis $e_{p,q}$, representing ordered translations around the two marked cycle families with a fixed scalar lift. Set

$$e_{p,q}e_{p',q'} = \omega^{q \cdot p'} e_{p+p', q+q'}, \quad e_{p,q}^* = \omega^{p \cdot q} e_{-p, -q}.$$

Extend the product bilinearly and the involution conjugate-linearly; $e_{0,0}$ is the identity. The commutator phase is the exponential of the signed intersection of the translation labels. These scalar crossing phases, not ordinary commutative multiplication of functions, define the quantized geometric multiplication.

Theorem 12.2 (Faithful and complete geometric matrix algebra). *In the orthonormal theta basis $|j\rangle$, put*

$$X_r|j\rangle = |j + \epsilon_r\rangle, \quad Z_r|j\rangle = \omega^{j_r}|j\rangle, \quad D_{p,q} = X^p Z^q.$$

Then $\mathfrak{Q}(e_{p,q}) = D_{p,q}$ is a unital $$ -algebra isomorphism*

$$\mathcal{A}_{g,k} \cong \text{End}(\mathcal{Q}_k), \quad T = \frac{1}{d} \sum_{p,q} \text{Tr}(D_{p,q}^\dagger T) D_{p,q}.$$

Every matrix has a unique reduced geometric symbol, preserving addition, composition and adjoint.

Proof. Theta translations give $Z_r X_s = \omega^{\delta_{rs}} X_s Z_r$, which proves the product and involution laws. A nonzero shift has zero trace; a nonconstant clock character sums to zero. Thus

$$\text{Tr}(D_{p,q}^\dagger D_{p',q'}) = d \delta_{p,p'} \delta_{q,q'}.$$

The $k^{2g} = d^2$ images form an orthogonal basis of the matrix algebra, proving injectivity, surjectivity and the inverse formula. Associativity also follows, or can be checked from the scalar cocycle. $\square$

Corollary 12.3 (States, observables and Born probabilities). *Define $\text{tr}_G(a) = d a_{0,0}$. Density operators correspond to positive symbols of trace one; positivity means $a = b^* b$ for some b . Pure state rays correspond to $a = a^* = a^2$ with trace one. Self-adjoint symbols are observables. Expectations and measurement probabilities are $\text{tr}_G(ao)$ and $\text{tr}_G(ae)$, for an effect $0 \leq e \leq 1$.*

Proof. The isomorphism preserves adjoints, products and trace. Positive matrices have square roots; trace-one projections have rank one. Transport these facts through $\mathfrak{Q}$. $\square$

Geometric multiplication can be evaluated without matrix multiplication:

$$(a \star b)_{r,s} = \sum_{p,q} a_{p,q} b_{r-p, s-q} \omega^{q \cdot (r-p)}.$$

This finite twisted convolution is the computational operation to validate; matrices are independent oracles.

Theorem 12.4 (Tensor and operator-vector compatibility). *For declared factor markings and theta structures,*

$$\mathcal{A}_{g_1+g_2,k} \cong \mathcal{A}_{g_1,k} \otimes \mathcal{A}_{g_2,k}, \quad D_{(p_1,p_2),(q_1,q_2)} = D_{p_1,q_1} \otimes D_{p_2,q_2}.$$

Output-major vectorization takes operators to $\mathcal{Q}_{\text{out}} \otimes \overline{\mathcal{Q}_{\text{in}}}$. The d^2 vectors $|D_{p,q}\rangle/\sqrt{d}$ form an orthonormal operator-vector basis. Contraction of the dual intermediate factors of $|T_2\rangle \otimes |T_1\rangle$ yields $|T_2 T_1\rangle$.

Proof. Cycle labels and crossing exponents split across factors; theta factorization supplies the Hilbert-space identification. Vectorization has inner product $\text{Tr}(T^\dagger S)$. Contracting intermediate indices gives $\sum_j (T_2)_{ij} (T_1)_{j\ell}$. Rectangular operators use matrix-unit symbols between distinct registers, not a square Weyl basis of the wrong dimension. $\square$

The contraction identity uses unnormalized operator vectors. If vectors are normalized as states, retain their norm factors; separately renormalizing each intermediate contraction would discard amplitude and success-probability information. For a connected surface the tensor labels are a declared register identification, not a claim of disjoint union; a coupled period matrix need not factor its theta functions. The decorated-cylinder interpretation uses linear combinations of curves modulo crossing/lift relations.

13 The modular action and the Weil representation

Proposition 13.1 (Period-to-modular coordinate bridge). *Let an orientation-preserving period map act on (a, b) by $M_{\text{per}} = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$, and suppose the output polarization is the corresponding change of marking. Then*

$$\Omega' = (R + S\Omega)(P + Q\Omega)^{-1}, \quad z' = (P + Q\Omega)^{-\top} z.$$

Its standard Siegel matrix is $M = \begin{pmatrix} S & R \\ Q & P \end{pmatrix}$.

Proof. The old holomorphic period blocks (I, Ω) become $(P + Q\Omega, R + S\Omega)$; renormalize the first block. Substituting $a' = Pa + Qb$, $b' = Ra + Sb$ in $b' - \Omega'a'$ and using symplecticity gives $(S - \Omega'Q)(b - \Omega a)$, with $S - \Omega'Q = (P + Q\Omega)^{-\top}$. $\square$

This is a change-of-marking theorem, not a claim that an arbitrary cobordism's output metric is the transported input metric. In genus one, $\tau' = (M_{21} + M_{22}\tau)/(M_{11} + M_{12}\tau)$. Never feed raw period M_{per} into a standard Siegel formula without this coordinate bridge. Below M always denotes the Siegel matrix.

A symplectic integer matrix $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \text{Sp}(2g, \mathbb{Z})$ acts on the Siegel space and on z by

$$\Omega' = (A\Omega + B)(C\Omega + D)^{-1}, \quad z' = (C\Omega + D)^{-\top} z.$$

Theorem 13.2 (Transformation law). *For k even, the theta space is carried to itself:*

$$\theta_j(z'; \Omega') = j(M, \Omega) \exp(\pi i k z'^\top (C\Omega + D)^{-1} C z) \sum_l \rho_k(M)_{jl} \theta_l(z; \Omega),$$

with $\rho_k(M)$ a $k^g \times k^g$ matrix independent of z and of Ω , unitary, and satisfying $\rho_k(M_1 M_2) = (\text{phase}) \rho_k(M_1) \rho_k(M_2)$. This is the Weil representation.

Here $j(M, \Omega)$ includes $\det(C\Omega + D)^{1/2}$ and a compatible metaplectic phase; it is not independent of Ω . For genus-one S , choose $j(S, \tau) = (-i\tau)^{1/2}$. The fiber norm and lift convention are part of the normalized identification. Odd levels require compatible characteristics or a restricted modular group; the displayed even-level generator formulas do not apply unchanged. The minimal fixed-basis odd-level implementation uses the theta subgroup: in genus one test S and T^2 , and in higher genus only generators preserving the characteristic (a shear with even diagonal is allowed). Reject T at odd level in that mode. Alternatively implement and track the transformed characteristics and line bundles explicitly.

Two generators suffice for $g = 1$. For $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ($\tau \mapsto \tau + 1$) the sum changes by $e^{\pi i k(n+j/k)^2}$, and for k even this is $e^{\pi i j^2/k}$, independent of n :

$$\rho_k(T) = \text{diag}(e^{\pi i j^2/k})_j, \quad \rho_2(T) = \text{diag}(1, i).$$

For $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ ($\tau \mapsto -1/\tau$) Poisson summation gives

$$\rho_k(S)_{jl} = \frac{1}{\sqrt{k}} e^{-2\pi i j l/k}, \quad \rho_2(S) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

the Hadamard gate. In the code $\rho_k(\mathbf{M})$ is never typed in: it is fitted from the transformation law at sampled z . The residual is a numerical check, not a proof of closure for all z ; use held-out samples and truncation convergence. The formulas above are independent checks.

A *shear* $\begin{pmatrix} I & B \\ 0 & I \end{pmatrix}$ with B symmetric integer sends $\Omega \mapsto \Omega + B$ and acts diagonally by $e^{\pi i j^\top B j/k}$. For $g = 2$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $k = 2$: the phase is $(-1)^{j_1 j_2}$, the controlled- Z gate.

Proposition 13.3 (Products). $\rho_k(\mathbf{M}_1 \oplus \mathbf{M}_2) = \rho_k(\mathbf{M}_1) \otimes \rho_k(\mathbf{M}_2)$. A *block-diagonal monodromy* is a product operator (operator Schmidt rank one) and cannot entangle; the shear above has Schmidt rank two.

Because $\rho_k(\mathbf{M})$ does not depend on Ω , the metric cannot deform it: on this register “the metric changes the representative, not the operator” holds verbatim (different τ are projectively equivalent polarizations under the declared projectively flat identification).

Proposition 13.4 (Why compatible geometric transport is unitary). A *compatible scalar lift of a symplectic cycle relabelling acts on the finite Heisenberg group preserving its center and adjoint. Its action on the theta register is implemented by a unitary operator unique up to phase. Compatible successive lifts compose projectively.*

Proof. The clock operators have a common eigenvector; its shift orbit gives the d distinct joint eigencharacters and spans an irreducible representation. The same construction in the relabelled representation gives an intertwiner U , unique up to scalar because the translations span the full matrix algebra. Preservation of adjoint implies $U^\dagger U$ commutes with every translation, hence is a positive scalar identity. Normalize that scalar to obtain a unitary. Composing intertwiners gives another intertwiner, so differs only by phase. The scalar lift and characteristic compatibility are hypotheses, especially at odd level. $\square$

14 Orientation reversal

An orientation-reversing period map satisfies $M_{\text{per}}^\top J M_{\text{per}} = -J$. Put $R_{\text{per}} = \text{diag}(I, -I)$ and $N = M_{\text{per}} R_{\text{per}}$, so N is symplectic. First apply R_{per} to the period coordinates $(a, b) \mapsto (a, -b)$:

$$\Omega \mapsto -\bar{\Omega}, \quad z \mapsto -\bar{z}, \quad \theta_j(-\bar{z}; -\bar{\Omega}) = \overline{\theta_j(z; \Omega)}.$$

Then apply the orientation-preserving bridge of Proposition 13.1 to N . With $\mathbf{N}$ its Siegel matrix the value-column action is $\rho_k(\mathbf{N})K$, where K is complex conjugation. The antiunitary lift is part of the data. For the torus period swap, $N = S$ and $\mathbf{N} = S^{-1}$, giving Hadamard followed by conjugation at level two. An antiunitary is not a CP channel; orientation reversal alone does not derive physical time reversal.

15 Coherent states and their exact covariance

Definition 15.1 (Evaluation coherent state). In the normalized theta basis, set $F_\Omega(z) = (\theta_j(z; \Omega))_j$ and

$$c_\Omega(z) = \frac{\overline{F_\Omega(z)}}{\|F_\Omega(z)\|_2}.$$

This is the Riesz representative of evaluation, up to its local line-fiber scalar. Its ray is well defined wherever evaluation is nonzero; reject a zero evaluation rather than divide by zero. A change of local line trivialization multiplies the vector by a phase after normalization.

Proposition 15.2 (Covariance with the value-column convention). *If the theta transformation law is $F_{\Omega'}(z') = f(z, \Omega) \rho_k(\mathbf{M}) F_\Omega(z)$ with unitary $\rho_k(\mathbf{M})$ and $f \neq 0$, then*

$$c_{\Omega'}(z') = \frac{\bar{f}}{|f|} \overline{\rho_k(\mathbf{M})} c_\Omega(z).$$

For the antiunitary value action $\rho_k(\mathbf{N})K$, the coherent action is $\overline{\rho_k(\mathbf{N})}K$, with the same scalar convention.

Proof. Conjugate the value-column law and divide by its norm. Unitarity removes the matrix from the norm; only $|f|$ remains. $\square$

One can instead define the state representation to be the conjugate representation from the outset. Either convention works; fitting the value representation and applying it unaltered to conjugated evaluations does not.

For a general flat cobordism, the period relation gives $(a_{\text{out}}, b_{\text{out}}) = M_{\text{per}}(a_{\text{in}}, b_{\text{in}})$ modulo integers. This alone gives no relation between Ω_{out} and Ω_{in} . Measure the holonomy relation separately. Different-modulus identity collars are negative controls: projectively flat identification of coefficient spaces is not a theorem that their evaluation states coincide.

Theorem 15.3 (Complete single-qubit coherent encoding). *For any elliptic curve E and positive degree-two line bundle L^2 , the coherent-state map $E \rightarrow \mathbb{CP}^1$ is onto. Thus every pure qubit ray is encoded by some real holonomy pair, at fixed polarization.*

Proof. Riemann–Roch on a genus-one curve gives $h^0(L^2) = 2$ and $h^0(L^2(-p)) = 1$ for every point p . Thus the two basis sections have no common zero. Their ratio defines a nonconstant holomorphic map $E \rightarrow \mathbb{CP}^1$; compactness and the open mapping theorem make its image all of $\mathbb{CP}^1$. Conjugating coordinates preserves surjectivity. The map has degree two, not a unique inverse. $\square$

This proves existence of an encoding, not physical preparation or a spontaneous choice of state. At fixed polarization a single elliptic curve cannot cover all qutrit rays in $\mathbb{CP}^2$. More generally the coherent family has complex dimension at most g , compared with $k^g - 1$ for all rays. Arbitrary state projectors remain representable by Corollary 12.3; coherent evaluation is only one family.

16 Seams, factorization and entangling capacity

Take two collars W_1, W_2 (each $T^2 \times [0, 1]$) and join them.

Proposition 16.1 (Sphere joins cannot entangle). *If the join is along a 2-sphere (remove a ball from each and identify the boundary spheres — a connected sum), then $H^1(W) = H^1(W_1) \oplus H^1(W_2)$, the monodromy is block-diagonal $M_1 \oplus M_2$, and $\rho_k(M) = \rho_k(M_1) \otimes \rho_k(M_2)$ for every choice of M_1, M_2 .*

Proof. Mayer–Vietoris with $W_1 \cap W_2 \simeq S^2$: $H^0(W_1) \oplus H^0(W_2) \rightarrow H^0(S^2)$ is onto and $H^1(S^2) = 0$, so $H^1(W) \rightarrow H^1(W_1) \oplus H^1(W_2)$ is an isomorphism. The external period relation therefore splits with the original markings, even if harmonic representatives have support in both pieces. Proposition 13.3 finishes it. $\square$

For valid gluing with overlap retracting to S^1 , Mayer–Vietoris imposes equality of restricted periods in $H^1(S^1; \mathbb{C})$. This may couple classes, but guarantees neither a full-domain graph, an invertible symplectic monodromy nor an entangling gate. An ordinary 3-dimensional 1-handle does not automatically provide this overlap; specify its attaching regions and verify the manifold and boundary topology. Mixing factors is not sufficient: SWAP has nontrivial operator Schmidt rank but maps every product state to a product state. Controlled- Z does entangle, sending $|+\rangle|+\rangle$ to $(|00\rangle + |01\rangle + |10\rangle - |11\rangle)/2$, with reduced state $I/2$. This witness distinguishes entangling capacity from operator-vector correlations.

Remark 16.2 (Measured seams). Section 35.2 records three constructions. A 1-handle between the far tori of two collars (a tube over one triangle of each) makes the far boundary one genus-two surface; its monodromy stays block-diagonal, while the surface’s own period matrix acquires an off-diagonal entry Ω_{12} set by the neck’s waist and length, and sampled coherent states have nonzero entropy in the declared two-factor register. Layered flip passes on a collar’s far torus read the Dehn twists T_A, T_B and the quarter turn S , and stacked passes compose in order. The collar of the $\mathbb{Z}/10$ -symmetric genus-two surface relabelled by its order-five rotation reads an integer symplectic monodromy of order five that acts on the level-two register as an entangling two-qubit Clifford gate, by the product-input witness above. A Dehn twist along a curve through the tube’s neck — the controlled- Z shear on a torus-pair register — remains open.

17 Clifford, and its limit

Weil operators normalize the finite Heisenberg–Weyl group and are Clifford operators. State explicitly which translations and central phases are included in a chosen lifted image; do not assume that it is a proper subgroup of the projective Clifford group. At even level lift and congruence conventions can involve $2k$, so naive identification with $\mathrm{Sp}(2g, \mathbb{Z}/k)$ loses phase data. At $k = 2$, these operators with stabilizer preparations and measurements are efficiently simulable and not quantum-universal. Metric-dependent transfer is not automatically a non-Clifford unitary.

The finite gate-group restriction does not contradict Theorem 12.2: the complex linear span of geometric translations is the entire matrix algebra, while products of a restricted set of unitary gates need not be dense.

18 From a Lie algebra to its root system

Everything non-abelian below is generated by one piece of structure: the eigenvalues of a maximal commuting family inside a Lie algebra. Fix a compact simple Lie algebra $\mathfrak{g}$ ($\mathfrak{su}(2)$, $\mathfrak{su}(3)$, ...) and a maximal abelian subalgebra $\mathfrak{h} \subset \mathfrak{g}$, the *Cartan subalgebra*; its dimension r is the *rank*. Complexify. The operators $\mathrm{ad} h = [h, \cdot]$, $h \in \mathfrak{h}$, commute and are diagonalizable, so

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{h}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_{\alpha}, \quad [h, x] = \alpha(h)x \quad (x \in \mathfrak{g}_{\alpha}),$$

with $\Phi \subset \mathfrak{h}_{\mathbb{C}}^*$ a finite set, the *roots*; each $\mathfrak{g}_{\alpha}$ is one-dimensional and $\Phi = -\Phi$. The roots are the weights of the adjoint representation. The Killing form $\kappa(x, y) = \mathrm{Tr}(\mathrm{ad} x \mathrm{ad} y)$ is nondegenerate

on $\mathfrak{h}$. On the real span of roots in the complexified dual use the induced positive Euclidean form (the compact real form has negative Killing form), normalized throughout so that long roots have $|\alpha|^2 = 2$. The *coroot* is $\alpha^\vee = 2\alpha/|\alpha|^2$. A generic hyperplane splits $\Phi = \Phi^+ \sqcup (-\Phi^+)$; the *simple roots* $\alpha_1, \dots, \alpha_r$ are the positive roots that are not sums of two positive roots, every positive root is a non-negative integer combination of them, and the *Cartan matrix* $A_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle$ has $A_{ii} = 2$ and $A_{ij} \in \{0, -1, -2, -3\}$ off the diagonal. The *highest root* θ is the positive root with the largest coefficients; it is long, so $\theta^\vee = \theta$.

Three lattices in $\mathfrak{h}^*$:

$$\begin{aligned} Q &= \mathbb{Z}\Phi && (\text{root lattice}), \\ Q^\vee &= \mathbb{Z}\Phi^\vee && (\text{coroot lattice}), \\ P &= \{\mu : \langle \mu, \alpha^\vee \rangle \in \mathbb{Z} \ \forall \alpha \in \Phi\} = (Q^\vee)^* && (\text{weight lattice}). \end{aligned}$$

P is spanned by the *fundamental weights* ω_i , $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$, so $\alpha_i = \sum_j A_{ij} \omega_j$ and $Q \subset P$ has index $\det A$, the order of the centre of the simply connected group. The *Weyl vector* is $\rho = \frac{1}{2} \sum_{\alpha > 0} \alpha = \sum_i \omega_i$ and the *dual Coxeter number* is $h^\vee = \langle \rho, \theta^\vee \rangle + 1$.

Example 18.1 ($\mathfrak{su}(2)$; Problem P48). $\mathfrak{g}_\mathbb{C} = \mathfrak{sl}(2, \mathbb{C})$ with $h = \text{diag}(1, -1)$, $e = E_{12}$, $f = E_{21}$: $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. One positive root α , $\alpha(h) = 2$, $\mathfrak{g}_\alpha = \mathbb{C}e$; $\alpha^\vee = h$; the fundamental weight has $\omega(h) = 1$, so $\alpha = 2\omega$, $P = \mathbb{Z}\omega$, $Q = 2\mathbb{Z}\omega$, $P/Q = \mathbb{Z}/2$, $|\omega|^2 = \frac{1}{2}$, $\rho = \omega$, $\theta = \alpha$, $h^\vee = 2$, $A = (2)$, $W = \{\pm 1\}$.

Example 18.2 ($\mathfrak{su}(3)$; Problem P49). $\mathfrak{sl}(3, \mathbb{C})$ with $\mathfrak{h}$ the traceless diagonal matrices and $\varepsilon_i(\text{diag}(a_1, a_2, a_3)) = a_i$. The roots are $\varepsilon_i - \varepsilon_j$, $i \neq j$, with root vectors E_{ij} : six roots. Simple roots $\alpha_1 = \varepsilon_1 - \varepsilon_2$, $\alpha_2 = \varepsilon_2 - \varepsilon_3$; the third positive root $\theta = \alpha_1 + \alpha_2$ is the highest. The Cartan matrix is $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$: the simple roots have length $\sqrt{2}$ at 120° , and the six roots are the vertices of a regular hexagon — the root system A_2 , whose root lattice is the hexagonal lattice. $\omega_1 = \frac{1}{3}(2\alpha_1 + \alpha_2)$, $\omega_2 = \frac{1}{3}(\alpha_1 + 2\alpha_2)$, $P/Q = \mathbb{Z}/3$ (triality), $\rho = \omega_1 + \omega_2 = \theta$, $h^\vee = 3$. The Weyl group permutes the ε_i : $W = S_3 \cong D_3$.

Proposition 18.3 (An internal A_2 lattice from scalar lengths). *Let a Euclidean lattice $Q = \mathbb{Z}\alpha_1 + \mathbb{Z}\alpha_2$ have a generating triangle with side lengths $\sqrt{2}, \sqrt{2}, \sqrt{6}$ at $0, \alpha_1, \alpha_2$. Its Gram matrix, recovered by the cosine law, is*

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The shortest nonzero lattice vectors are $\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2)$; they form A_2 . Their reflections preserve Q , generate S_3 , and the metric-dual lattice is $P = A^{-1}\mathbb{Z}^2$ in root coordinates, with $|P/Q| = 3$.

Proof. $\alpha_1 \cdot \alpha_2 = (2 + 2 - 6)/2 = -1$. The squared norm of $m\alpha_1 + n\alpha_2$ is $2(m^2 - mn + n^2)$, whose smallest positive value is two at exactly the six stated integer pairs. Reflection in α_i is $v \mapsto v - (v \cdot \alpha_i)\alpha_i$; the integral Gram entries prove lattice preservation, and the two reflections have product of order three. Duality gives P and $\det A = 3$. $\square$

This internal lattice is chosen independently of the spacetime boundary (Section 32); a generic boundary triangulation need not have this symmetry, nor its harmonic mass Gram equal A .

Theorem 18.4 (Highest-weight classification). *The finite-dimensional irreducible representations of $\mathfrak{g}$ are V_λ , one for each dominant weight $\lambda \in P^+ = \{\lambda \in P : \langle \lambda, \alpha_i^\vee \rangle \geq 0 \ \forall i\}$. $V_\lambda = \bigoplus_{\mu \in P} V_\lambda[\mu]$ decomposes into weight spaces, h acting on $V_\lambda[\mu]$ by $\mu(h)$; λ occurs once, every other weight is λ minus a non-negative combination of simple roots, and the weights counted with multiplicity form a W -invariant multiset.*

For $\mathfrak{su}(2)$, $V_\lambda = \text{Sym}^\lambda \mathbb{C}^2$ (spin $\lambda/2$) has weights $\lambda, \lambda - 2, \dots, -\lambda$ in units of ω , each once: dimension $\lambda + 1$. The *character* $\text{ch } V_\lambda = \sum_\mu \dim V_\lambda[\mu] e^\mu$ is a function on the maximal torus $T = \exp \mathfrak{h}$ through $e^\mu(\exp h) = e^{\mu(h)}$; it is exactly the object of Section 8: characters multiply under $\otimes$ because weights add.

19 The Weyl group and the character formula

The reflection in the root α is $s_\alpha(\mu) = \mu - \langle \mu, \alpha^\vee \rangle \alpha$. The *Weyl group* $W = \langle s_\alpha : \alpha \in \Phi \rangle$ is finite, generated by the simple reflections $s_i = s_{\alpha_i}$, permutes Φ , preserves P , Q , $Q^\vee$ and $\langle \cdot, \cdot \rangle$, and has the closed dominant chamber $\{\langle \mu, \alpha_i^\vee \rangle \geq 0\}$ as a fundamental domain. Its sign character is $\varepsilon(w) = \det w = (-1)^{\ell(w)}$.

Theorem 19.1 (Weyl character formula; Problem P52). *For $\lambda \in P^+$,*

$$\text{ch } V_\lambda = \frac{\sum_{w \in W} \varepsilon(w) e^{w(\lambda + \rho)}}{\sum_{w \in W} \varepsilon(w) e^{w\rho}}, \quad \sum_{w \in W} \varepsilon(w) e^{w\rho} = e^\rho \prod_{\alpha > 0} (1 - e^{-\alpha}),$$

and $\dim V_\lambda = \prod_{\alpha > 0} \langle \lambda + \rho, \alpha^\vee \rangle / \langle \rho, \alpha^\vee \rangle$.

For $\mathfrak{su}(2)$, writing $x = e^\omega$: $\text{ch } V_\lambda = (x^{\lambda+1} - x^{-\lambda-1}) / (x - x^{-1}) = x^\lambda + x^{\lambda-2} + \dots + x^{-\lambda}$. The numerator is the *antisymmetrization* of the single exponential $e^{\lambda+\rho}$ under W ; every formula below has that shape.

Proposition 19.2 (Brauer–Klimyk; weights add, the Weyl group folds; Problem P53). *Let $m_\mu(\nu) = \dim V_\mu[\nu]$. Then, as virtual representations,*

$$V_\lambda \otimes V_\mu = \sum_{\nu \in P} m_\mu(\nu) \varepsilon(w_\nu) V_{w_\nu(\lambda + \nu + \rho) - \rho},$$

where $w_\nu \in W$ takes $\lambda + \nu + \rho$ into the dominant chamber and the term is omitted when $\lambda + \nu + \rho$ lies on a wall of the chamber. The right side has non-negative coefficients.

Proof. Multiply the numerator of $\text{ch } V_\lambda$ by $\text{ch } V_\mu = \sum_\nu m_\mu(\nu) e^\nu$. Since m_μ is W -invariant, $\sum_w \varepsilon(w) e^{w(\lambda + \rho)} \sum_\nu m_\mu(\nu) e^{w\nu} = \sum_\nu m_\mu(\nu) \sum_w \varepsilon(w) e^{w(\lambda + \nu + \rho)}$, and each inner sum is $\varepsilon(w_\nu)$ times the numerator of $\text{ch } V_{w_\nu(\lambda + \nu + \rho) - \rho}$, or zero when $\lambda + \nu + \rho$ is fixed by a reflection. Divide by the denominator. $\square$

For $\mathfrak{su}(2)$: $V_1 \otimes V_\lambda = V_{\lambda+1} \oplus V_{\lambda-1}$ for $\lambda \geq 1$ (Clebsch–Gordan), and $V_1 \otimes V_0 = V_1$ because $\lambda + \nu + \rho = 0$ for $\nu = -1$ lies on the wall. This is the finite prototype of the Kac–Walton rule of Section 23: at level k one more wall appears and one more reflection folds across it.

20 Affine Lie algebras: the level, the translations, and theta functions

The register needs the loop version of $\mathfrak{g}$. The *affine Lie algebra* is

$$\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K \oplus \mathbb{C}d, \quad [x \otimes t^m, y \otimes t^n] = [x, y] \otimes t^{m+n} + m \delta_{m+n,0} \langle x, y \rangle K,$$

with K central and $d = t \frac{d}{dt}$ the grading. A representation on which K acts by the scalar k has *level* k . The Cartan subalgebra grows to $\hat{\mathfrak{h}} = \mathfrak{h} \oplus \mathbb{C}K \oplus \mathbb{C}d$, a weight is $\hat{\lambda} = (\lambda, k, n)$, and there is one new simple root $\alpha_0 = \delta - \theta$, where δ generates the imaginary roots ($\delta(d) = 1$, $\delta|_{\mathfrak{h} \oplus \mathbb{C}K} = 0$). Dominance for α_0 reads $\langle \hat{\lambda}, \alpha_0^\vee \rangle = k - \langle \lambda, \theta^\vee \rangle \geq 0$:

Theorem 20.1 (Kac). *The integrable highest-weight modules $L(\hat{\lambda})$ of level k are indexed by the Weyl alcove $P_k^+ = \{\lambda \in P^+ : \langle \lambda, \theta^\vee \rangle \leq k\}$.*

The affine Weyl group is $\hat{W} = W \ltimes t(Q^\vee)$: the finite Weyl group together with translations t_γ , $\gamma \in Q^\vee$, which act on a weight of level K by

$$t_\gamma(\hat{\lambda}) = \hat{\lambda} + K\gamma - (\langle \lambda, \gamma \rangle + \frac{K}{2}|\gamma|^2)\delta,$$

i.e. by $\lambda \mapsto \lambda + K\gamma$ on the finite part. It is generated by $s_1, \dots, s_r$ and the affine reflection $s_0 = t_{\theta^\vee} s_\theta$, the reflection in the hyperplane $\langle \mu, \theta^\vee \rangle = K$ (Problem P55). At level K the fundamental alcove $\{\mu : \langle \mu, \alpha_i^\vee \rangle \geq 0, \langle \mu, \theta^\vee \rangle \leq K\}$ is a simplex and a fundamental domain, and $P_k^+ + \rho$ is its set of interior lattice points at $K = k + h^\vee$.

Theorem 20.2 (Weyl–Kac; the lattice sum is the translation orbit; Problem P56). *With $\hat{\rho} = (\rho, h^\vee, 0)$,*

$$\text{ch } L(\hat{\lambda}) = \frac{\sum_{\hat{w} \in \hat{W}} \varepsilon(\hat{w}) e^{\hat{w}(\hat{\lambda} + \hat{\rho}) - \hat{\rho}}}{\prod_{\hat{\alpha} > 0} (1 - e^{-\hat{\alpha}})^{\text{mult } \hat{\alpha}}}.$$

Write $q = e^{-\delta} = e^{2\pi i \tau}$ and $e^\mu = e^{2\pi i \langle \mu, z \rangle}$, and split $\hat{W} = W \ltimes t(Q^\vee)$. The translation orbit of a single exponential at level K is a theta function,

$$\sum_{\gamma \in Q^\vee} e^{t_\gamma(\hat{\mu})} = q^{-|\mu|^2/2K} \Theta_\mu^{(K)}(\tau, z), \quad \Theta_\mu^{(K)}(\tau, z) = \sum_{\gamma \in Q^\vee} q^{\frac{K}{2}|\frac{\mu}{K} + \gamma|^2} e^{2\pi i K \langle \frac{\mu}{K} + \gamma, z \rangle},$$

so, with $A_\mu^{(K)} = \sum_{w \in W} \varepsilon(w) \Theta_{w\mu}^{(K)}$, the numerator at $K = k + h^\vee$ is $A_{\lambda+\rho}^{(K)}$ up to a power of q , the denominator is $A_\rho^{(h^\vee)}$ (the affine denominator identity), and the normalized level- k character with highest weight $\lambda \in P_k^+$ is

$$\chi_\lambda(\tau, z) = \frac{A_{\lambda+\rho}^{(k+h^\vee)}(\tau, z)}{A_\rho^{(h^\vee)}(\tau, z)}.$$

This is the structural content of “geometry is algebra” in the non-abelian register. The translations $t(Q^\vee)$ are a lattice, and the lattice sum they generate is a theta function — the object Section 10 already attaches to a torus. The finite Weyl group W antisymmetrizes the sum, but does not replace the root/coroot lattices, their normalization or the level. An abelian theta register has trivial Weyl group and no Weyl denominator; substituting $h^\vee = 0$ into the non-abelian character ratio would leave its denominator undefined.

Theorem 20.3 (Sugawara; Problem P57). *On a level- k integrable module the Virasoro generator $L_0 = \frac{1}{2(k+h^\vee)} \sum_a :J^a J^a:$ acts on the highest-weight vector of $L(\hat{\lambda})$ by the conformal weight $h_\lambda = C_2(\lambda)/2(k+h^\vee)$, $C_2(\lambda) = \langle \lambda, \lambda + 2\rho \rangle = |\lambda + \rho|^2 - |\rho|^2$, with central charge $c = k \dim \mathfrak{g}/(k+h^\vee)$. The T -matrix of the modular data is $T_\lambda = e^{2\pi i(h_\lambda - c/24)}$.*

With the strange formula $|\rho|^2/2h^\vee = \dim \mathfrak{g}/24$ this is $T_\lambda = \exp 2\pi i[|\lambda + \rho|^2/2(k+h^\vee) - \dim \mathfrak{g}/24]$, the form quoted in Theorem 23.1. For $\mathfrak{su}(2)_k$: $h_\lambda = \lambda(\lambda+2)/4(k+2)$ (spin $j = \lambda/2$: $j(j+1)/(k+2)$) and $c = 3k/(k+2)$; $c = 1$ at $k = 1$, $c = \frac{3}{2}$ at $k = 2$. This has Ising-type fusion, but is not the standard Ising conformal theory, whose central charge is $1/2$; twists also distinguish them.

21 Root systems of rank two

A boundary torus supplies a rank-two integral lattice with an alternating intersection form, not a positive root Gram or a Lie algebra. If a rank-two Euclidean root system is additionally chosen, the possibilities are $A_1 \times A_1$, A_2 , B_2 , G_2 . The complex dimension of the base elliptic curve differs from the rank of the Cartan fiber; affine characters on a torus can have a Lie algebra of any rank. With long roots of length $\sqrt{2}$ and α_1 long (Problem P50):

	$A_1 \times A_1$	A_2	B_2	G_2
Cartan A	$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$	$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$	$\begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$	$\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$
Gram $\langle \alpha_i, \alpha_j \rangle$	$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$	$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$	$\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$	$\begin{pmatrix} 2 & -1 \\ -1 & \frac{2}{3} \end{pmatrix}$
root lattice	square	hexagonal	square	hexagonal
W	D_2 (4)	D_3 (6)	D_4 (8)	D_6 (12)
P/Q	$\mathbb{Z}/2 \times \mathbb{Z}/2$	$\mathbb{Z}/3$	$\mathbb{Z}/2$	1
$h^\vee$	(2, 2)	3	3	4

$W(A_1 \times A_1) \subset W(B_2)$ and $W(A_2) \subset W(G_2)$ have index two. For $A_1 \times A_1$, levels and dual Coxeter numbers are specified per simple factor. The abelian case $\mathfrak{g} = \mathfrak{u}(1)$ has no roots, trivial W , $P = \mathbb{Z}$: the theta functions of Section 10 are the $r = 1$, $W = 1$ abelian construction, with no non-abelian denominator.

Proposition 21.1 (Rank-two Weyl groups are the symmetries a torus can have; Problem P59). *The finite subgroups of $\mathrm{GL}(2, \mathbb{Z})$ are the crystallographic point groups: cyclic of order 1, 2, 3, 4, 6 and dihedral D_1, D_2, D_3, D_4, D_6 . The dihedral ones D_2, D_3, D_4, D_6 are the four rank-two Weyl groups. A finite group of simplicial automorphisms of a torus induces such a subgroup on H_1 . Translations can lie in the kernel, so the full simplicial automorphism group need not embed in $\mathrm{GL}(2, \mathbb{Z})$.*

The square grid with one diagonal realizes only $\{\pm I, \pm \sigma\} \cong D_2$ by simplicial automorphisms — the diagonal breaks the fourfold symmetry — which is abelian, so composition order is invisible among automorphisms; the layered flip passes of Section 35.2 are not automorphisms and read the infinite-order twists T_A , T_B and the quarter turn S on the same grid. The triangular (hexagonal) lattice realizes $D_6 = W(G_2)$, non-abelian, containing $D_3 = W(A_2)$.

22 The alcove and the Verlinde space

At level k the analogue of the characteristics $j \in \mathbb{Z}/k$ is the *Weyl alcove* of Theorem 20.1,

$$P_k^+ = \{\lambda \in P^+ : \langle \lambda, \theta^\vee \rangle \leq k\},$$

the dominant weights of level at most k . For $\mathfrak{su}(2)$: $\lambda = 0, 1, \dots, k$ (spin $0, \frac{1}{2}, \dots, \frac{k}{2}$), so $k + 1$ states. The torus Hilbert space is spanned by one function per alcove weight: the *Weyl–Kac character* of the level- k integrable representation of the affine algebra $\hat{\mathfrak{g}}$.

By Theorem 20.2 the character is the ratio $\chi_\lambda = A_{\lambda+\rho}^{(k+h^\vee)} / A_\rho^{(h^\vee)}$: the sum over $Q^\vee$ is the translation part of $\hat{W}$, the sum over W its finite part (Problems P62, P63). For $\mathfrak{su}(2)$ ($r = 1$, $W = \{\pm 1\}$, $h^\vee = 2$) this is fully explicit in the theta functions of Definition 10.1 (with the variable z rescaled by the root normalization):

$$\chi_\lambda(\tau, z) = \frac{\theta_{\lambda+1}^{(2(k+2))}(z; \tau) - \theta_{-(\lambda+1)}^{(2(k+2))}(z; \tau)}{\theta_1^{(4)}(z; \tau) - \theta_{-1}^{(4)}(z; \tau)}, \quad \lambda = 0, \dots, k,$$

where $\theta_m^{(K)}$ is the level- K theta function of characteristic $m \bmod K$. The register's construction is: root data $\rightarrow$ alcove $\rightarrow$ antisymmetrize the existing lattice sum under $W \rightarrow$ divide by the denominator. A disjoint union of tori is still a tensor product, with basis (λ_1, λ_2) .

22.1 Coordinates and polarization of the root-lattice register

For the simply-laced controls A_1, A_2 , let A be the coroot Gram matrix, write the Cartan variable as ξ in root coordinates, and let $m \in \mathbb{Z}^r$ be the Dynkin coordinates of μ . Then

$$\mu = A^{-1}m, \quad c = \frac{A^{-1}m}{K}, \quad \Omega = \tau A, \quad z_\theta = A\xi.$$

The generalized theta sum is

$$\Theta_\mu^{(K)}(\tau, \xi) = \sum_{n \in \mathbb{Z}^r} e^{\pi i K(n+c)^\top \tau A(n+c) + 2\pi i K(n+c)^\top A \xi}.$$

Characteristics are indexed by $P/KQ^\vee$, not arbitrary unrelated rational points. The physical Cartan torus in these coordinates has $\xi \sim \xi + n + \tau n'$ with polarization KA . Consequently its section count is $\det(A)K^r$: $2K$ for A_1 , $3K^2$ for A_2 . These are not the K^r sections of the principal genus- r bundle. Using $\Omega = \tau A$ is a convenient series representation, not a license to replace the period lattice or its polarization.

Under $\tau \mapsto -1/\tau$, Poisson summation over $Q^\vee$ mixes its dual-lattice cosets. This is not the ordinary genus- r Siegel inversion: $(\tau A) \mapsto -(\tau A)^{-1}$ gives $-A^{-1}/\tau$, not $-A/\tau$. Under $\tau \mapsto \tau + 1$, Ω changes by A , not by I . At S , the numerator at level $K = k + h^\vee$ and denominator at level $h^\vee$ have common determinant factors and their exponential quotient is $\exp(\pi i k \xi^\top A \xi / \tau)$. Derive and fit this root-lattice law directly. Evenness of A_1, A_2 permits integer K ; the odd-level restriction of the principal rank-one register is a different issue.

For A_2 the surviving character dimension is $|P_k^+| = (k+1)(k+2)/2$, hence three at $k=1$ and six at $k=2$. The ratios are normalized affine characters $\text{Tr}_{L_\lambda}(q^{L_0 - c/24} e^{2\pi i(\cdot, \xi)})$. On a Weyl wall numerator and denominator may both vanish: use a controlled removable limit or evaluate away from the walls with a denominator bound. Do not divide numerical zeros.

23 Kac–Peterson, Verlinde, and multiplication into addition

Theorem 23.1 (Kac–Peterson). *The characters transform under $\text{SL}(2, \mathbb{Z})$ among themselves, $\chi_\lambda(-1/\tau, z/\tau) = (\cdots) \sum_\mu S_{\lambda\mu} \chi_\mu(\tau, z)$ and $\chi_\lambda(\tau + 1, z) = T_{\lambda\lambda} \chi_\lambda(\tau, z)$, with*

$$S_{\lambda\mu} = c \sum_{w \in W} \varepsilon(w) \exp\left(-\frac{2\pi i}{k+h^\vee} \langle w(\lambda + \rho), \mu + \rho \rangle\right), \quad T_{\lambda\lambda} = \exp\left(2\pi i \left[\frac{|\lambda + \rho|^2}{2(k+h^\vee)} - \frac{\dim \mathfrak{g}}{24}\right]\right),$$

with $c = i^{|\Phi^+|} |P/(k+h^\vee)Q^\vee|^{-1/2}$ and T the Sugawara matrix of Theorem 20.3 (Problems P64, P57). For $\mathfrak{su}(2)_k$: $S_{\lambda\mu} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{\pi(\lambda+1)(\mu+1)}{k+2}\right)$ and $T_\lambda = \exp\left(2\pi i \left[\frac{(\lambda+1)^2}{4(k+2)} - \frac{1}{8}\right]\right)$.

These replace the abelian Weil matrices. A numerical character experiment can fit their transformation laws with held-out residuals, controlling lattice truncation and removable denominator zeros. Fitting geometric lattice sums against independent character formulas validates the representation objective.

Theorem 23.2 (Verlinde). *The fusion coefficients $N_{\lambda\mu}^\nu$ — the multiplicities in the level- k truncated tensor product — are*

$$N_{\lambda\mu}^\nu = \sum_{\sigma \in P_k^+} \frac{S_{\lambda\sigma} S_{\mu\sigma} \overline{S_{\nu\sigma}}}{S_{0\sigma}},$$

and are non-negative integers. For $\mathfrak{su}(2)_k$, $N_{\lambda\mu}^\nu = 1$ exactly when $|\lambda - \mu| \leq \nu \leq \min(\lambda + \mu, 2k - \lambda - \mu)$ and $\lambda + \mu + \nu$ is even, else 0.

The Kac–Walton formula computes the same numbers by *adding* the weights of V_μ to λ and reflecting every result back into the alcove by the affine Weyl group, with the sign of the reflection. That is the precise sense of “root systems turn multiplication into addition”: characters multiply, weights add, and the Weyl group folds. It is Proposition 19.2 with $\hat{W}$ in place of W : the one new ingredient is the affine reflection s_0 (Problem P65).

Definition 23.3 (Fusion algebra). Let $\mathcal{V}_k$ have basis f_λ , $\lambda \in P_k^+$, and product $f_\lambda f_\mu = \sum_\nu N_{\lambda\mu}^\nu f_\nu$. This fixed-level fusion product is not ordinary pointwise multiplication of affine characters at a common arbitrary (τ, ξ) ; such a product does not preserve the level.

Theorem 23.4 (Fusion as multiplication on geometric alcove points). *For the unitary modular data above, put*

$$v_\lambda(\sigma) = \frac{S_{\lambda\sigma}}{S_{0\sigma}}, \quad \sigma \in P_k^+.$$

Then $f_\lambda \mapsto v_\lambda$ is an algebra isomorphism $\mathcal{V}_k \otimes \mathbb{C} \cong \mathbb{C}^{P_k^+}$ with pointwise multiplication. Moreover the Weyl formula identifies

$$v_\lambda(\sigma) = \text{ch } V_\lambda \left(\exp \left[-\frac{2\pi i(\sigma + \rho)}{k + h^\vee} \right] \right).$$

Thus fusion can be calculated by evaluating finite characters at specified points of the internal geometric torus, multiplying values, and changing back to the alcove basis.

Proof. The last identity is the ratio of the two finite Weyl alternating sums in $S_{\lambda\sigma}$ and $S_{0\sigma}$. The evaluation matrix is invertible because S is unitary and $S_{0\sigma} > 0$. Substituting the Verlinde formula and summing over ν using unitarity gives $\sum_\nu N_{\lambda\mu}^\nu v_\nu(\sigma) = v_\lambda(\sigma) v_\mu(\sigma)$. The unit is $v_0 = 1$. $\square$

The representation is exact; independent Kac–Walton folding checks the integral nonnegative structure constants.

24 Simply-laced level one and the abelian lattice register

Theorem 24.1 (Frenkel–Kac–Segal; Problem P58). *Let $\mathfrak{g}$ be simply laced (A, D, E : all roots of length $\sqrt{2}$, $Q = Q^\vee$). The level-one modules $L(\hat{\omega}_i)$, with ω_i ranging over 0 and the minuscule fundamental weights, are realized on the Fock space of r free bosons tensored with $\mathbb{C}[Q + \omega_i]$, and their characters are the theta functions of the cosets of the root lattice in the weight lattice:*

$$\chi_{\omega_i}(\tau, z) = \frac{\Theta_{Q+\omega_i}(\tau, z)}{\eta(\tau)^r}, \quad \Theta_{Q+\omega}(\tau, z) = \sum_{\gamma \in Q+\omega} q^{|\gamma|^2/2} e^{2\pi i \langle \gamma, z \rangle},$$

where $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$.

Since η transforms by a multiplier under $\mathrm{SL}(2, \mathbb{Z})$, the modular data of $\mathfrak{g}$ at level one is that of the *even lattice* Q with characteristics P/Q : $S_{ab} = |P/Q|^{-1/2} e^{-2\pi i \langle a, b \rangle}$ and $T_a = e^{-2\pi i r/24} e^{\pi i |a|^2}$ for $a, b \in P/Q$ (Poisson summation over Q , whose dual is P). For $\mathfrak{su}(N)_1$, $P/Q = \mathbb{Z}/N$, $S_{ab} = N^{-1/2} e^{2\pi i ab/N}$ and $T_a \propto e^{\pi i a(N-a)/N}$. For $\mathfrak{su}(2)_1$: $Q = \sqrt{2} \mathbb{Z}$, the two coset theta functions are the level-two functions $\theta_0^{(2)}, \theta_1^{(2)}$ of Definition 10.1, S is the Hadamard matrix and $T \propto \mathrm{diag}(1, i)$: the abelian register at $k = 2$ is $\mathfrak{su}(2)$ at level one. A diagonal T exists here, and not for the rank-one register at odd level (Section 13), because Q is even: $|\gamma|^2/2 \in \mathbb{Z}$ on Q , so $\tau \mapsto \tau + 1$ multiplies each coset sum by the constant $e^{\pi i |\omega_i|^2}$. The rank-one register at level k is the lattice $\sqrt{k} \mathbb{Z}$, even exactly when k is even. In particular $\mathfrak{su}(3)_1$ has relative twists $(1, e^{2\pi i/3}, e^{2\pi i/3})$ from the A_2 discriminant quadratic form. Matching the group $\mathbb{Z}/3$ alone, or using the principal rank-one level-three register, is not the Frenkel–Kac comparison.

25 Finite on the torus, dense with punctures

Two facts fix what a non-abelian register can and cannot do.

Theorem 25.1 (Congruence property; Ng–Schauenburg). *For a modular tensor category the projective modular representation on the unpunctured torus has a congruence kernel and finite projective image. Exact level and linear-lift statements require fixing the normalization of T and its anomaly phases.*

So on a single torus the non-abelian register, like the Clifford one, gives a finite set of operators.

Theorem 25.2 (Density in specified Jones sectors; Freedman–Larsen–Wang). *Certain Jones braid representations have projectively dense image in the corresponding unitary group. Level, strand labels, total fusion charge and irreducible sector must satisfy the density theorem’s hypotheses. This is not a statement for every sector of every punctured sphere.*

Punctures and braids are the proposed route. Removing a bulk top cell does not by itself construct a labeled surface puncture or conformal-block Hilbert space, and the Weyl isotypic sectors of an operator are not alcove labels. Section 32 states what a realization must supply.

Part II

Experiment specification

26 Whole harmonic states and identifiable operators

This section states the native harmonic-recovery contract. Its three verdicts are identifiable operator, positive norm preservation, and requested native gate realization; they are separate from the representation verdict of Section 30. A harmonic space is a candidate state space; a nonzero vector or ray needs input data or a geometric selection rule. For positive Hermitian masses,

$$\langle h, L_p h \rangle_{M_p} = \|\delta_p h\|_{M_{p-1}}^2 + \|d_p h\|_{M_{p+1}}^2.$$

This proves closedness and coclosedness of the kernel. Orthogonality of exact, harmonic and coexact summands gives a unique representative per class. Complex bilinear masses do not provide this positivity proof. For a flat torus with $\tau = a + ib$, $b > 0$,

$$g_\tau = \frac{1}{b} \begin{pmatrix} 1 & a \\ a & a^2 + b^2 \end{pmatrix}, \quad \omega_\tau = dx + \tau dy, \quad *\omega_\tau = -i\omega_\tau.$$

Periods give $\psi = (1, \tau)/\sqrt{1 + |\tau|^2}$ under a declared identity metric on period coordinates, not assumed Whitney orthonormality. This fixed-orientation positive-real construction covers one Bloch hemisphere; complex continuation needs separate diagnostics.

26.1 Frozen whole-kernel recovery

Assemble $K(\lambda) = \tilde{A} - \lambda M_{\text{whole}}$ and let Z span its geometric zero kernel in the coordinates on which readouts act. Set $A = R_{\text{in}} Z$, $B = R_{\text{out}} Z$.

Proposition 26.1 (Identifiability). *A unique operator T on every input with $B = TA$ exists exactly when A is onto and $\ker A \subseteq \ker B$.*

Proof. Necessity is immediate. Choose a right inverse C of A . Since $c - CAc \in \ker A$, $Bc = BCAC$; hence $T = BC$ works. Right inverses differ by a map into $\ker A$, so give the same T . Onto-ness gives uniqueness. The equivalent uniqueness test is $B(I - CA) = 0$. $\square$

Square invertible A gives $T = BA^{-1}$. An unchecked pseudoinverse is not a certificate. Invisible modes are allowed if both readouts annihilate them. The zero map is valid but has no normalized Choi ray. Record thresholds, ranks, singular gaps, conditioning and full-space residuals. For basis and fixed-seed held-out inputs x , reconstruct h and independently check

$$R_{\text{in}} h = x, \quad \tilde{A} h = 0, \quad R_{\text{out}} h = Tx.$$

A minimum Euclidean image-norm witness resolves invisible modes computationally, not physically. Frozen recovery accepts only geometry and declared readouts, never targets or desired outputs. Rebasings Z leaves T unchanged; port rebasing gives $T' = g_{\text{out}}^{-1} T g_{\text{in}}$ with transformed port metrics.

26.2 Chain/cochain and zero-mode certificates

Distinguish the native chain frame Φ from cochain images $Z = G^U \Phi$. For twisted boundary operators ∂_j^U , test

$$(\partial_2^{U^{-1}})^\top Z = 0, \quad \partial_1^U \Phi = 0.$$

These are closedness on images and coclosedness represented on the chain frame. Compare an independent null space with the band's rank/span; require finite projector idempotency, rank tolerance, singular-gap and resolvent certificates. Distinguish contour rank, algebraic multiplicity, geometric kernel and closed/coclosed dimension on defective pencils. Check the map to cohomology, not only equality with the Betti number. A mismatch is a failed hypothesis, not permission to repair geometry. Flat scalar local systems support twisted cohomology, pairing U with U^{-1} . Nontrivial flat $U(1)$ holonomy need not be pure gauge; forbidding matrix connections does not remove it. Curved U can have $d_U^2 \neq 0$, so cohomological gluing needs new justification.

27 Pairing and Gram conventions

Reuse the live marked block's native dual/primal frames $F^\vee, F$, normalized on its own Whitney mass, then embed the dual with E :

$$(F^\vee)^\top M_{\text{own}} F = I, \quad R_{\text{Gram}} \Phi_W = (E F^\vee)^\top M_{\text{whole}} \Phi_W.$$

Reuse `PencilSchur.gramBlock`; do not duplicate mass assembly or dual normalization. This bilinear observation is not an orthogonal projection; transpose is intentional. Periods use cochain images and remain a distinct topological control. For boundary harmonic chain bases $\Phi, \Phi^\vee$, let $P, P^\vee$ be periods of their images and $B_0 = (\Phi^\vee)^\top M_{\text{own}} \Phi$. Square invertible period matrices give

$$G_{\text{per}} = (P^\vee)^{-\top} B_0 P^{-1}.$$

The abbreviated $(P^\vee)^{-\top} P^{-1}$ requires $B_0 = I$. Zero-connection primal and dual/primal constructions agree only when they describe the same period-normalized pairing; primal-frame and biorthogonal observation coordinates are distinct. With primal/dual transports $M, M^\vee$, pairing preservation tests $(M^\vee)^\top G_B M = G_A$. Using M on both sides assumes coincident transports in the declared trivialization. This is distinct from positive quantum isometry. A complex bilinear form is not a Krein inner product, which must be nondegenerate indefinite Hermitian.

28 Internal elimination and actual gluing

For retained/internal coordinates b, z , invertible $K_{zz}(0)$ gives

$$z = -K_{zz}^{-1} K_{zb} b, \quad S_G = K_{bb} - K_{bz} K_{zz}^{-1} K_{zb}, \quad S_G b = 0.$$

This Schur complement is exact elimination, not a partial trace; internal coordinates remain reconstructable. Singular blocks require compatibility and null modes, not an unchecked pseudoinverse. The positive Hermitian fixture $S = [-T, I]^\dagger W [-T, I]$, $W > 0$, has zero energy exactly for $y = Tx$. For two unit-weight constraints put $y = T_1 x + u$, $r = z - T_2 T_1 x$. Minimizing $\|u\|^2 + \|r - T_2 u\|^2$ gives $u = (I + T_2^\dagger T_2)^{-1} T_2^\dagger r$ and energy

$$r^\dagger (I + T_2 T_2^\dagger)^{-1} r.$$

The zero relation composes to $z = T_2 T_1 x$. A target-built fixture is an algebraic control, not scalar-geometric synthesis. Actual composition must identify interface cells/frames and compare the assembled glued relation. Extra dimension does not by itself give matrix realizability: a connected positive scalar degree-zero connection graph has at most one zero mode, since $f_j = e^{i\theta_{ij}} f_i$ fixes all values from one vertex and inconsistent holonomy forces zero.

29 Quantum norms, tensor products and channels

Declare positive Hermitian $Q_{\text{in}}, Q_{\text{out}}$ and test $T^\dagger Q_{\text{out}} T = Q_{\text{in}}$. Orthonormal coordinates use $\hat{T} = Q_{\text{out}}^{1/2} T Q_{\text{in}}^{-1/2}$. Only with a declared positive norm can a harmonic vector be normalized as a quantum state. Stationary-action relaxation does not prove isometry. Fix output-major vectorization:

$$|T\rangle\rangle = \sum_{i,j} \hat{T}_{ij} |i\rangle_{\text{out}} \otimes |j\rangle_{\text{ref}}, \quad (I \otimes \langle \bar{x} |) |T\rangle\rangle = \hat{T} x.$$

The reference is the conjugate/dual input, not another physical boundary. For nonzero $\hat{T}$, normalize by $\|\hat{T}\|_F$. Schmidt coefficients are normalized singular values; the reference marginal is $(\hat{T}^\dagger \hat{T})^\top / \|\hat{T}\|_F^2$, equal to I/d_{in} for an isometry. This is operator-vector information, not automatically physical-state entanglement. For Kraus operators in orthonormal coordinates,

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger, \quad J_{\mathcal{E}} = \sum_a |K_a\rangle\rangle \langle\langle K_a|.$$

CP is $J_{\mathcal{E}} \geq 0$; TP is $\text{Tr}_{\text{out}} J_{\mathcal{E}} = I_{\text{in}}$ in this unnormalized convention. Stinespring requires $V : A \rightarrow B \otimes E$, $V^\dagger V = I$, and $\mathcal{E}(\rho) = \text{Tr}_E(V \rho V^\dagger)$. A third boundary in $B \oplus E$ is not that tensor product.

29.1 Three distinct tensor constructions

1. Künneth over a field: $H^n(X \times Y) \cong \bigoplus_{p+q=n} H^p(X) \otimes H^q(Y)$. An n -fold degree-one register occupies multidegree $(1, \dots, 1)$, not generally all of H^n . Harmonic factorization needs a compatible product complex/metric. Künneth is not limited to quantization level one.
2. Theta quantization gives k^g sections from a rank- $2g$ lattice with principal polarization and level line bundle; it is not H^1 . The modulus register uses τ to select a line in H^1 ; the theta register uses τ as polarization and a section as state.
3. Ordinary root-system tensor multiplicities come from Brauer–Klimyk; affine fusion comes from Verlinde/Kac–Walton. Fusion is not the untruncated tensor product. Disjoint surfaces factor state spaces; sewing/punctures require sums over labels and fusion spaces.

Two level-two theta factors give four amplitudes, a 4×4 gate and sixteen operator-vector amplitudes. A direct sum of two harmonic port frames, or the 2×2 selected-pair χ , is not that operator. Dimension matching alone does not define subsystems.

30 Representation acceptance contract

The deliverable is a representation theorem with computational checks of its hypotheses and operations, not a dynamics or synthesis claim. Theorems hold exactly under their assumptions; numerical residuals measure implementation accuracy. Each objective identifies which operations are geometric computations and which are independent matrix oracles.

- V1. State the classical/quantum type bridge, integral markings, positive polarization, level, theta structure and coefficient/evaluation convention. Use positive Hodge masses for the classical proof, or report a certified alternative; do not import positivity from a complex bilinear pencil.
- V2. Geometric operator algebra: represent $e_{p,q}$ by scalar translations; implement reduced cycle/translation symbols, scalar crossing phases, involution, trace and twisted convolution. Verify the crossing law, involution, the d^2 independent symbols, the trace pairing, forward and inverse maps for arbitrary operators, adjoints and multiplication against independent matrices, including non-Clifford and nonunitary operators. Keep the pinned-boundary defect as a distinct invariant control, not an optimization target; add no relaxation to the fixed-complex verifier.
- V3. Holonomy-defined states: keep the neutral harmonic pencil, construct ordinary closed phase configurations and read Wilson loops independently; verify gauge and large-gauge invariance and neutral-pencil independence. Fix $z = b - \Omega a$, the coordinate bridge, the conjugate coherent action and the antiunitary factorization. Demonstrate qubit coverage with several target rays including zeros and poles, and its non-unique inverse. At level three use a named characteristic-preserving subgroup or track characteristics explicitly; do not claim coherent coverage of every qutrit ray.
- V4. Tensor and Choi algebra: verify tensor-factor ordering, matrix units, symbol tensor products and dual-factor contraction against composition, including rectangular maps, product, SWAP and controlled- Z controls, an entangling witness and the state/channel distinction. Keep the wider-neck geometry study (Section 33) a separate representation and entropy experiment with paired samples, period accuracy and refinement, and logarithmic entropy asymptotics; it is neither a requirement for the algebra theorem nor a claim about Ryu–Takayanagi.
- V5. Root-lattice representation: recover A_2 , its roots, reflections and the weight lattice from scalar lattice lengths; use the nonprincipal bundle and the $P/KQ^\vee$ characteristic set of Section 22.1. Fit normalized Weyl–Kac characters against Kac–Peterson on held-out points; compare fusion by geometric alcove evaluation, Verlinde and independent Kac–Walton folding, with A_1 and the even-lattice Frenkel–Kac controls. Host-derived τ is an input source; the native fiber need not be synthesized.
- V6. Use fixed-seed held-out inputs distinct from fit samples, independent formulas, conditioning/rank reports and truncation/precision convergence. Report residuals before normalization/rounding and reject failed hypotheses. Surface periods require symmetry, positive imaginary part, and a justified complex structure; report $J^2 + I$ and any normalization rather than hiding it. Validate refinement where a discrete read approximates continuum geometry.
- V7. Keep default `verify/theta` fixed-complex and read-only with respect to engine evolution. Reuse existing Gram/pencil/theta implementations. Preserve compatibility except for explicitly corrected scientific expectations; add regressions demonstrating why they changed.
- V8. Each record declares its `certifies` scope: exact algebra representation, sampled analytic-family check, approximate geometric readout, or native realization. A native realization failure is not a representation failure. No matrix connection, engine relaxation, universality, speedup or entropy–area law is required or inferred.

31 Native implementation and acceptance contract

These requirements govern the native-engine path where it is exercised; they do not replace V1–V8. The verifier’s check names (R1–R5, H1–H3, J1–J3, C1, D1, U1, F1–F4, G1–G5, E1–E6, T1–T9) are a separate namespace from the contract items below. Incorrect scientific claims are corrected with regressions, not preserved as acceptance criteria.

31.1 Construction NC1–NC7

- NC1. A native harmonic state is a whole cochain including boundary; restrictions are readouts. A theta-register state is instead a section, or its density symbol; these are different types.
- NC2. Reuse Whitney pencils, `MultiCobordism`, fibers and shared `emergence_animation.py` runtime; keep `qubit_animation.py` canonical.
- NC3. Preserve pinned defaults and `--no-pin-boundary`; never remove boundary cells.
- NC4. Preserve manifold/collar gates, stage-1 search and stage-2 scalar length/phase relaxation. No matrix-valued connection or target in links, measured frames or assembled Laplacians.
- NC5. Targets may condition synthesis, never frozen recovery.
- NC6. Preserve complex/Lorentzian geometry; do not silently impose quantum positivity.
- NC7. Read the zero kernel, distinguish excited/generalized bands and apply Section 26.

31.2 Scientific readout NR1–NR7

- NR1. Retain flat-torus inputs, markings, holomorphic fibers and own-state residuals.
- NR2. Retain scalar collar synthesis and shared engine-unit schedule.
- NR3. Record the actual objective: transfer/ χ fitting optimizes selected-state amplitudes, not the whole operator.
- NR4. Recover whole-kernel relations each frame through periods and native dual-frame whole-metric contractions using `PencilSchur.gramBlock` and Section 27. The `gram` record specifies `observation_convention: dual_frame_whitney`. Records without this field used primal frames; normalizations need not agree even at zero phase. Identity coordinate norms are explicit external encodings.
- NR5. Preserve χ , not as a full XY/two-qubit gate and never as input to frozen recovery.
- NR6. Publish `correspondence`: identifiability, ranks, conditioning, basis/held-out errors, norm preservation, Choi data and a separate gate verdict.
- NR7. Report the rank-two modulus host’s obstruction to a four-dimensional two-qubit gate despite objective convergence. Theta sphere joins give product operators; entanglement needs a validated operator and product-input witness, not merely a proposed circle seam.

31.3 Compatibility NK1–NK5

- NK1. Preserve native collar/bridge gates and rollback.
- NK2. Preserve own-surface frames and block-state residuals.
- NK3. Keep bilinear pencil transfer distinct from whole transport and certified channels.
- NK4. Keep readout in `qubit_animation.py` with reusable helpers; preserve CLI, geometry records, output handling, neutral runtime and the χ measurements while displaying whole-operator evidence.
- NK5. `verify/theta` read fixed complexes seeded through `seed_qubit_host`, without node, objective, readout optimization or engine stages. `MultiCobordism.restriction` supplies one basis, all marking periods, chain frame and certificate; `monodromy` adds its fit. Preserve parameter/modulus-named records under `~/cobordism-runs/functor-verify/` and `~/cobordism-runs/theta-register/`, with a delimiting `certifies` sentence. `run.sh` runs both.

31.4 Acceptance NA1–NA9

- NA1. Live collar: kernel residual, input coverage, unique output and basis/held-out whole-cochain witnesses; no target in recovery.
- NA2. Independent complex, rectangular, hidden-mode, ambiguous-output, incomplete-coverage, zero-map and rank-loss fixtures; rebase invariance.
- NA3. Choi contractions, marginals, norms and composition; no unitary certificate for nonunitary or zero maps.
- NA4. Perturb scalar geometry: periods stay topological on fixed marked collars, Gram observations may move. For pure gauge $g(v) = e^{i\phi(v)}$ fixed to one at port basepoints, both operators remain unchanged for real and complex lengths. Otherwise the native convention gives $T' = e^{-i(\phi_{\text{out}} - \phi_{\text{in}})}T$; normalized Choi marginals, entropy and Schmidt coefficients remain invariant. Retain $\mathbb{C}^*$ dual gauge-covariance controls.
- NA5. Shared headless/live records, finite-safe JSON, panels naming claims/obstructions and CLI/runtime regressions.
- NA6. A failed gate realization remains failed when selected-state fitting converges.
- NA7. Preserve `verify` at 10^{-12} on product/swap two-/four-torus and identical-square hosts: rank, integrality, declared isotropy, flipped control, closed and coclosed independent kernels and certificates before and after jitter, fixed periods, exact representative/Gram motion, twist-group law, direct sum and transport defect. Keep doubled-swap, $(1, 10^{-5})$ exactness, isometric-start, 10^{-7} rebase, same-convention zero-phase Gram and $U(1)/\mathbb{C}^*$ gauge regressions; report counts and failures.
- NA8. Preserve `theta` T1–T8: transformation law, norm, projective law, generators, factorization, antiunitarity and controlled- Z . Record level, truncation, fit/held-out samples, phases and factor ordering. Algebraic generator controls do not prove native host synthesis.

NA9. Preserve the tube, flip and decagon reads (Section 35.2): `verify --far-flips F1–F4 and theta --far-flips T8/T9` (the Dehn twists and composition order on one host, grid 3 for one pass, grid 5 for two); `theta --tori 4 --join tube G1–G5` (the genus-two far boundary: product monodromy, period matrix, coupling, geometric-state entropy, neck sweep); `theta --surface decagon E1–E6` (the entangling monodromy of the order-five rotation, its fixed period matrix, the Clifford gate, the product-collar control). Report counts and failures; records under `~/cobordism-runs/`.

32 Root representation and optional native synthesis

Native fiber synthesis is optional. The internal coroot lattice is independent of boundary cohomology: boundary mapping classes act on $H^1(\Sigma; \mathbb{Z})$, whereas the Weyl group acts on $Q^\vee$. Even when the abstract groups coincide, these are not the same action.

For a torus boundary the abelianized Cartan phase lattice is $H^1(\Sigma; \mathbb{Z}) \otimes Q^\vee$, with the product of the surface intersection form and the positive internal root form, followed by the declared affine reduction. The surface supplies τ ; internal scalar lattice geometry can supply A_2 , its roots, weights and reflections. The series notation $\Omega = \tau \text{Gram}(Q^\vee)$ must retain the nonprincipal polarization and characteristic set of Section 22.1. A genus- r boundary whose principal period matrix happens to equal τA is not, by that equality alone, the root register.

The acceptance claim (V5) is that the lattice/alcove construction reproduces the character and fusion algebra. Native synthesis would add a sector map, certified whole-kernel readouts and actual sewing; braiding would add labeled punctures and compatible associativity and braiding maps.

33 Neck geometry and encoded-state entropy

Definition 33.1 (Entropy in a declared tensor register). For a normalized pure section coefficient vector in $\mathcal{Q}_1 \otimes \mathcal{Q}_2$, let p_i be its squared Schmidt coefficients. Its reduced von Neumann entropy in bits is $S = -\sum_{p_i > 0} p_i \log_2 p_i$. This measures the encoded state relative to the declared factors, not an intrinsic function of one period entry.

Proposition 33.2 (Weak-coupling entropy is generally not quadratic). *Suppose a two-qubit normalized state varies analytically with a real parameter ϵ , is product at zero, and has a nonzero first-order component normal to the product-state manifold. Its small Schmidt probability obeys $p = C\epsilon^2 + O(\epsilon^3)$, $C > 0$, and*

$$S = -C\epsilon^2 \log_2(C\epsilon^2) + \frac{C\epsilon^2}{\ln 2} + o(\epsilon^2).$$

For level-two theta states with $\Omega = \text{diag}(\tau_1, \tau_2) + \epsilon \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ at $z = 0$, the first derivative vanishes. If the second-order entangling component is nonzero, $p \sim C\epsilon^4$ instead.

Proof. The small singular value is first order in the normal component, so its square is $C\epsilon^2 + O(\epsilon^3)$. Expand the binary entropy $-p \log_2 p - (1-p) \log_2(1-p)$. For the theta family, $\partial_\epsilon \theta|_0 = (2\pi i k)^{-1} \theta'_{j_1}(z_1; \tau_1) \theta'_{j_2}(z_2; \tau_2)$. Both level-two sections are even functions of their argument, so their first derivatives vanish at zero. The leading entangling amplitude can then be second order; special parameters may cancel it as well. $\square$

The star-tube experiment uses wider disk attachment, waist/length sweeps and larger grids, with $|\Omega_{12}| \sim 0.3\text{--}0.5$ as an exploration target. Record full Ω , markings, diagonal drift, period-reader

errors, mesh resolution and the same fixed real holonomy samples (a, b) at each point; convert to $z = b - \Omega a$. Synthetic asymptotic controls at fixed z, τ_1, τ_2 are different from geometry sweeps in which these quantities move. Report both Schmidt probabilities and entropy; fit the appropriate logarithmic asymptotics, separating $z = 0$ from generic points. Topology/positivity at one strong point does not establish convergence. A physical degeneration must be checked on actual geometry; manually zeroing Ω_{12} is only a factorization control.

This is encoded-state entanglement, not a Ryu–Takayanagi validation. No independent extremal-area observable, area normalization or holographic dictionary is supplied.

34 Information and geometric diagnostics: unresolved definitions

Total-system entropy against engine units and curvature/charge heatmaps are not operator-vector entropy. Before reporting them, declare a positive normalized joint ρ , subsystem factors and an evolution or coarse-graining rule. Record $S(\rho) = -\text{Tr}(\rho \log \rho)$, reduced entropies and $I(A : B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$, with log base and engine schedule. A globally pure closed state has zero total entropy while local entropies may change; their sum is not total entropy. Mutual information measures total correlations, not an intersection of directional flows or entanglement alone. An MI-dependent inner product at the 3-simplex needs a joint-state construction and a Hermitian positive-semidefinite kernel for every finite set; symmetry/nonnegativity alone are insufficient. Nondegeneracy is additionally needed.

Unobserved harmonic remainder is not automatically an environment or lost entanglement. Retain it in reconstruction until projection, conditioning or partial trace is declared with normalization and CP/TP status. Transport/merging must propagate the joint state and correlations under that operation. Curvature heatmaps need an estimator, dual-cell support, Lorentzian/complex convention, units and normalization; separate *mass curvature* needs a defined mass observable and geometric relation. Charge maps need an observable, sign, support and conservation test.

Part III

Evidence, reconciliation and references

35 What the code measures

`qubit.animation.py verify` reads a seeded collar at tolerance 10^{-12} : rank and integrality of M (R1–R4); isotropy for the declared form on the orthonormalized restriction subspace with the flipped control nonvanishing (R5, Remark 3.3 and P6); the harmonic certificates (H1–H3, the caveat after Theorem 2.2); metric independence of M against an $O(1)$ motion of the representative that is exactly a coboundary (J1–J3, Theorem 2.4); the twist-group composition (C1, P14); the four-torus direct sum (D1, Proposition 16.1); and the covariant transport defect (U1, Section 7 and Remark 7.1). The subcommand `theta` fits ρ_2 for the generators and the seeded monodromies (T1–T8, Sections 13–16). These are fixture results, not proofs by numerical precision. Residuals lie at 10^{-16} – 10^{-14} ; metric-dependent defects can be order one or small for matching boundaries.

status	statement and limitation
theorem	L_W is Lagrangian for oriented manifold boundaries over a field; harmonic readout additionally needs a certified cohomology identification.
conditional	M is integral unimodular under integral lattice isomorphisms, not merely invertible period matrices (Theorem 4.2).
theorem	Relations compose by cohomological gluing; disjoint union gives direct sum. Symbol contraction is a separate exact composition theorem; native seam gluing requires independently assembled complexes.
theorem	Theta section spaces factor; even-level Weil operators are Clifford. This is a new register, not equality with harmonic H^1 .
measured	Bilinear transport defect 0.46–0.58 generically, about 10^{-14} for identical tori; conventions are in Section 27.
measured (Section 35.2)	Gluing along the far surface of one host by layered flip passes, with the Dehn twists T_A , T_B , S and their non-commuting composition; a genus-two far boundary whose period matrix couples the tori while the monodromy stays a product; an entangling monodromy from the order-five rotation of a symmetric genus-two surface.
open	Geometric operator algebra, holonomy/coherent encoding, tensor/Choi composition and neck entropy, root-lattice characters and fusion (V2–V5).
optional	Native seam and through-neck gate synthesis, positive physical dynamics and non-abelian fiber/sewing implementation; these do not gate the representation result.

35.1 Numerical evidence on the collar fixtures

For grid 3, seed 7 and tolerance 10^{-12} , **verify** and **theta** report:

- Period integrality residual at most 10^{-14} ; product/swap maps equal identity/swap; declared isotropy at most $1.1 \cdot 10^{-14}$, versus flipped controls 1.41 (two tori) and 2.00 (four).
- A 75% complex length jitter changes M by at most $1.6 \cdot 10^{-14}$ but moves the canonical representative by 0.49–0.77, with exact difference residual at most $4 \cdot 10^{-14}$. Reported Gram motion is 0.18–0.45.
- Twist-group composition residual at most $1.5 \cdot 10^{-14}$; sphere-join off-block norm at most $5 \cdot 10^{-15}$. The automorphism twists of the square triangulation commute; composition order is tested by the flip passes (F4, T9).
- Closed/coclosed residuals at most $2.5 \cdot 10^{-14}$, nullity/rank 2 or 4, independent span residual at most $1.2 \cdot 10^{-14}$ and projector idempotency at most $7 \cdot 10^{-16}$.
- Level-two theta-law residuals at most $6.8 \cdot 10^{-16}$; unitarity $1.3 \cdot 10^{-15}$; projective group law $9.5 \cdot 10^{-16}$; Hadamard/phase checks $4.4 \cdot 10^{-16}$; factorization $1.9 \cdot 10^{-16}$; Kronecker law $6.2 \cdot 10^{-16}$; controlled- Z $5.3 \cdot 10^{-16}$; polarization independence $8.8 \cdot 10^{-16}$. Product/CZ operator Schmidt ranks are one/two. The algebraic CZ control is not a native entangling host.
- Reported **verify** counts: 17/17 for each of product, swap and identical-square two-torus fixtures; 27/27 for each four-torus product/swap fixture. Reported **theta** counts: 8/8, 8/8, 10/10, 10/10 on two-/four-torus product/swap fixtures.

These fixed-complex reads establish no engine dynamics, interior-seam composition or non-abelian register; a triangulation-preserving search move can leave the marked relation unchanged.

35.2 Measurements on the tube, flip and decagon hosts

On the fixed complexes named, tolerance 10^{-12} , grid 3 and seed 7 unless stated; records under `~/cobordism-runs/functor-verify/` and `~/cobordism-runs/theta-register/`.

- **Tube-joined collars** (`theta --tori 4 --join tube`). Two collars joined by a prism tube between one far triangle of each: the far boundary is one genus-two surface, $\partial W = T^2 \sqcup T^2 \sqcup \Sigma_2$, Betti numbers $[1, 4, 2, 0]$, restriction rank $4 = b_1(\partial W)/2$, boundary Euler characteristics $[-2, 0, 0]$, declared isotropy $4.6 \cdot 10^{-15}$ against the flipped control 2.0 (G1). The monodromy from the near tori to Σ_2 is I_4 (product collars) or $\text{swap} \oplus \text{swap}$, off-block norm $2.6 \cdot 10^{-15}$, operator Schmidt rank one (G2): the tube alone couples no cycle. The surface's own Riemann period matrix, read from its harmonic 1-forms with the torus reader's conventions at genus two, has $\text{Im } \Omega$ positive definite (eigenvalues 0.83, 1.08), $\Omega_{12} = 1.45 \cdot 10^{-4} - 1.88 \cdot 10^{-4}i$ on the default neck and is symmetric to $2.4 \cdot 10^{-15}$ at every neck read (G3). The geometric (coherent) state at a holonomy point, in the basis the identity monodromy identifies with $\Theta_2 \otimes \Theta_2$, has entanglement entropy exactly 0 for the diagonal control ($5 \cdot 10^{-30}$) and $4 \cdot 10^{-7}$ bits mean on the default neck (G4). Over 25 necks (waist 0.25–4, length 0.25–4) $|\Omega_{12}|$ runs $5 \cdot 10^{-8}$ to $2.4 \cdot 10^{-2}$, falling approximately exponentially with tube length in the reported sweep. The entropy range is $5 \cdot 10^{-14}$ to $2.4 \cdot 10^{-3}$ bits mean; its proportionality to $|\Omega_{12}|^2$ over this range is a finite-range observation, not the asymptotic law. The diagonal of Ω approaches the far tori's moduli as the neck pinches ($0.099 \rightarrow 0.037$): the separating degeneration (G5). Counts 11/11, 11/11 (swap), 12/12 (sweep). This is a weak-coupling encoded-state observation, not an entropy–area law; the averaged values are not paired-sample convergence evidence. The asymptotics and controls are in Section 33.
- **Dehn twists by layered flips** (`verify --far-flips, theta --far-flips`). One tetrahedron per flipped boundary edge of the far torus (nine per pass on grid 3): rows for T_B , columns for T_A , diagonals then the quarter turn for S ; the flipped grid is the grid in a sheared basis, relabelled by the lattice map L , and the far marking reads $M = L^T$. Measured $T_B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $T_A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ with residual $\leq 4 \cdot 10^{-15}$, of infinite order for the twists (F2); the flipped far torus's own modulus in the relabelled marking equals $(M_{21} + M_{22}\tau)/(M_{11} + M_{12}\tau)$ to $8 \cdot 10^{-16}$ (F3); $\rho_2 = \text{diag}(1, i)$, $H \text{diag}(1, -i)H$, H at $\leq 5 \cdot 10^{-16}$ (T8). On grid 5: T_B^2 , $T_A T_B$ and $T_B T_A$ read as the products of their passes, $T_A T_B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $T_B T_A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ (F4), with Weil matrices at projective distance 1.0 (T9); with the swap, $M = M(\text{flips}) \cdot \text{swap}$. Counts 20/20 (one pass), 21/21 (two passes), 8/8, 9/9. On grid 3 a second pass collides with an existing edge and is refused by name.
- **The entangling collar** (`theta --surface decagon`). The decagon with opposite sides identified, coned and barycentrically subdivided (28 vertices, 90 edges, 60 faces, $\chi = -2$), the flat decagon metric, its rotations as vertex permutations (simplicial isometries of order ten), a symplectic marking by integer symplectic reduction of the intersection form read from the surface (cup pairing of the harmonic basis, sign calibrated on the flat torus). The collar relabelled by r^2 reads $M = \begin{pmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \end{pmatrix}$ on (A_1, A_2, B_1, B_2) as predicted from the side classes, $\det 1$, $M^T J M = J$, $M^5 = I$, mixing (E2); the surface's own period matrix is fixed by M to $1.5 \cdot 10^{-15}$, $\text{Im } \Omega$ with eigenvalues 0.46, 1.66 (E3); $\rho_2(M)$ unitary (residual $6 \cdot 10^{-16}$), $\rho^5 \propto I$

($1.5 \cdot 10^{-15}$), operator Schmidt rank four (E4); 16 of the 36 product stabilizer states are sent to Bell pairs (exactly one bit), projective distance 1.00 to every $A \otimes B$ and 0.77 to every $\text{SWAP} \cdot A \otimes B$ over the 24×24 single-qubit Cliffords (E5); the product collar reads $M = I$ with entropy $5 \cdot 10^{-29}$ (E6). r^4 : order five, Schmidt rank two, entangling; r : order ten, entangling; $r^5 = -I$: Schmidt rank one, no entangling power. Counts 13/13 each. Reduced mod 2 an element of order five of $\text{Sp}(4, \mathbb{F}_2) \cong S_6$ lies in no conjugate of $S_3 \wr \mathbb{Z}/2$, so no such gate is a local Clifford or a swap times one: the entangling capacity is forced by the order.

Modular conventions of the theta records. The theta records fit ρ from the period matrix directly (the direct-monodromy input convention). They certify the recorded fits and witnesses, not covariance with the simultaneously reported boundary modulus; covariance tests apply Propositions 13.1 and 15.2, distinguishing raw period matrices, standard Siegel matrices, and value/state actions. A discrete period read is not an exact Riemann-surface theorem by symmetry and positivity alone.

36 Where this sits

- Gelca–Uribe, [From classical theta functions to topological quantum field theory](#): scalar translation/crossing algebra, theta quantization and the linearized decorated-cobordism interpretation.
- Manoliu, [Abelian Chern–Simons theory](#): a scalar- $U(1)$ TQFT completion with extended-manifold gluing data. It is not derived from the engine’s Regge relaxation.
- Riemann–Roch for elliptic curves (e.g. Hartshorne, *Algebraic Geometry*, Ch. IV): the degree-two line-bundle argument in Theorem 15.3.
- Ryu–Takayanagi, [Holographic derivation of entanglement entropy from AdS/CFT](#): an extremal-area correspondence with additional physical hypotheses, not a name for a theta neck curve.
- Atiyah, *Topological quantum field theories* (1988): the tensor-monoidal axioms against which the direct-sum relation functor of Section 6 must be distinguished.
- Mumford, *Tata Lectures on Theta I*: Definition 10.1, the transformation law, and the Heisenberg group of Section 10.
- Weil, *Sur certains groupes d’opérateurs unitaires* (1964): the representation of Section 13.
- Gottesman, *The Heisenberg representation of quantum computers* (1998): why Clifford is classically simulable.
- Humphreys, *Introduction to Lie Algebras and Representation Theory*, Ch. II–III and VI: root systems, the Weyl group and the character formula of Sections 18–19.
- Fulton–Harris, *Representation Theory*, Lectures 11–13: $\mathfrak{sl}(2)$ and $\mathfrak{sl}(3)$ worked by hand.
- Kac, *Infinite dimensional Lie algebras*, Ch. 6 (the affine Weyl group), 10–13 (integrable modules, the character formula and the modular matrices of Sections 20–23).
- Frenkel–Kac, *Basic representations of affine Lie algebras and dual resonance models* (1980): Theorem 24.1.
- Goddard–Olive, *Kac–Moody and Virasoro algebras in relation to quantum physics* (1986): the Sugawara construction.
- Di Francesco, Mathieu, Sénéchal, *Conformal Field Theory*, Ch. 14–16: the $\mathfrak{su}(2)_k$ formulas, Verlinde, Kac–Walton.
- Ng–Schauenburg, *Congruence subgroups and generalized Frobenius–Schur indicators* (2010): [congruence and finiteness on the torus](#).
- Freedman–Larsen–Wang, *The two-eigenvalue problem and density of Jones representation of braid groups* (2002): [density with representation-specific hypotheses](#).
- NIST DLMF, [Section 21.5, modular transformations](#): the determinant factor and characteristics.
- Arnold–Falk–Winther, [Finite element exterior calculus](#): positive Hodge theory, not positivity of a complex Lorentzian pencil.
- Watrous, [Channel representations](#): Choi, Kraus and vectorization conventions.

- Implementation context:
simplicial_qubit_spec.md;
include/chainhodge/ChainHodge.h;
include/cobordism/PencilLayer.h;
include/cobordism/MultiCobordism.h;
tessera/quantum/theta_register.py.